

Appendix S1

False Positive Catamenial Epilepsy Classification: A Simulation Study

Development and calibration of HORMONE-CYCLE and supplementary study methods

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Chapter 1. Development of HORMONE-CYCLE

1.1 Purpose, scope, and intended use

HORMONE-CYCLE is a custom, open-source simulator developed for this project. It generates reproducible daily menstrual-cycle, bleeding, ovulation, estradiol, and progesterone diaries from age, diary length, a seed, and optional modifiers. (1) HORMONE-CYCLE is the complete software title. The simulator supports method development, operating-characteristic studies, and controlled synthetic examples. Its outputs represent synthetic trajectories. Individual clinical forecasting falls outside the intended use.

The design follows a hierarchical simulation workflow and reports inputs, data-generating mechanisms, estimands, and reproducibility artifacts. (2–4) Population evidence constrains age-specific distributions, each synthetic person receives stable latent traits, each cycle receives a stochastic realization, and a daily renderer produces the final diary.

1.2 Evidence review and role of each source

Three primary data sources supplied baseline calibration targets: the Apple Women’s Health Study for age-specific cycle summaries, the Natural Cycles cohort for phase and bleeding timing, and daily laboratory measurements for estradiol and progesterone subphase values. (5–7) Cunningham et al. supplied a fourth, independent aggregate cohort that was held out from parameter fitting and used only for a 12-month age-stratified cross-check. (8) Condition-specific sources informed modifier directions or broad ranges; exact modifier margins are investigator-selected regression guards. Table A1 states exactly how each source entered development and evaluation. Development used aggregate quantities reported in the articles; participant-level source data were unavailable and were not used.

Appendix Table A1. Evidence provenance for HORMONE-CYCLE. “Calibration” means that a published result informed parameter fitting, waveform construction, or an acceptance target. “Held out” means that the published summary was not used to fit or construct the relevant model component. Cunningham is an independent aggregate cycle source; Anckaert is an independent aggregate hormone source. Neither provides participant-level linked validation for the simulated joint trajectories.

Paper or evidence source	Population or data	Published result used	Use during development	Use during evaluation	Independence status
Li et al. (2023) (5)	165,668 cycles from 12,608 Apple Women’s Health Study participants	Age-specific mean cycle length, pooled within-person SD, participant-level irregularity, and short/long cycle tails	Direct or derived age-band outcome targets; surrogate initialization followed by full-simulator refinement of latent two-component variability parameters	Prespecified age-stratified calibration checks	Development source reused internally
Bull et al. (2019) (6)	612,613 ovulatory cycles from 124,648 Natural Cycles users	29.3-day cycle, 4.0-day bleeding, 16.9-day follicular, and 12.4-day luteal means in Table 1	Direct phase-timing and bleeding targets; stable-luteal structure	Phase and bleeding mean/SD checks	Development source reused internally
Stricker et al. (2006) (7)	Daily samples from 20 healthy volunteers, synchronized to the luteinizing-hormone peak	Complete daily serum estradiol and progesterone medians synchronized to the LH peak	Primary ovulatory daily-envelope construction after unit conversion and event alignment	Daily-envelope and prespecified morphology software checks	Development source reused for internal construction checks
Cunningham et al. (2024) (8)	>19 million global Flo users aged 18–55; 12-month analysis cohort	Age-specific participant cycle mean and mean participant-specific cycle-length SD	Not used for fitting	Held-out aggregate cross-check with prespecified practical-equivalence margins	Independent aggregate source; no participant-level linkage
Harlow et al. (2000) (9)	416 conception and non-conception cycles; 28 follicular phases lasting ≥ 24 days	Five heterogeneous long-follicular urinary-oestrogen patterns; delayed dominant-follicle emergence was most common	Qualitative geometry for delayed-emergence and failed-wave E2 variants; no class-frequency fit	Visual long-follicular morphology check	Development morphology source; 25% failed-wave share remains investigator selected

Paper or evidence source	Population or data	Published result used	Use during development	Use during evaluation	Independence status
Mumford et al. (2012) (10)	259 healthy regularly menstruating women aged 18–44; up to two cycles and eight serum samples per cycle	In cycles >35 days, E2 and LH peaks occurred about three days later than in normal-length cycles	Independent timing context supporting delayed rather than proportional stretching	Qualitative long-cycle timing cross-check	Independent context source; not used to estimate waveform values
Anckaert et al. (2021) (11)	85 apparently healthy women aged 18–37 monitored across one natural cycle	Assay-specific E2 and P4 follicular, ovulatory, luteal, and subphase reference intervals	Not used to construct the waveform	Independent amplitude and phase-order cross-check with broad assay-aware bounds	Independent aggregate hormone source; not participant-level trajectory validation
Fraser et al. (2011) (12)	International expert terminology and normal bleeding framework	Standardized bleeding terminology	Bleeding interpretation and bounds	Contextual review	Context
Mortimer et al. (2026); Doi et al. (2005) (13,14)	Longitudinal cycle-pattern evidence and a hormone/regularity study in polycystic ovary syndrome	Longer and more variable cycles with attenuated luteal progesterone	Initial polycystic-ovary-syndrome modifier directions; numerical multipliers are calibrated constants	Context for internal directional checks	Development sources reused internally
Mortimer et al. (2026); Jarrett et al. (2020) (13,15)	160,206 cycles from 15,586 participants; intensive ultrasound and hormone study	Age-dependent cycle differences and disordered follicle development in polycystic ovary syndrome	Later corroboration of the implemented profile	Directional scenario checks	Later corroboration
Venturoli et al. (1987) (16)	Adolescent hormonal-pattern and ovarian-morphology study	Irregularity and incomplete ovulation early after menarche	Initial early-postmenarche modifier directions; numerical constants are calibrated choices	Context for an internal stress test	Development source reused internally
World Health Organization study (1986); Zhang et al. (2008) (17,18)	Adolescent diary cohorts and a two-year daily hormone study	Longer, irregular, and incompletely ovulatory early-postmenarche cycles	Later corroboration of the implemented profile	Directional scenario checks	Later corroboration
Santoro and Randolph (2011) (19)	Review of menopause-transition endocrine evidence	Greater variability, more anovulation, and variable estradiol	Perimenopause modifier	Internal directional stress test	Development source reused internally
Edelman et al. (2014) (20)	Cochrane review of cyclic and continuous combined hormonal contraception	Ovulation suppression and regimen-specific bleeding patterns	Combined oral-contraceptive profiles	Internal directional and bounded stress tests	Development source reused internally
Xiao et al. (1995); Barbosa et al. (1995) (21,22)	Long-term levonorgestrel intrauterine-device hormone and ultrasound studies	Ovulation usually persists while bleeding and amenorrhea patterns change	Levonorgestrel intrauterine-device profile	Internal ovulation and amenorrhea checks	Development sources reused internally

Paper or evidence source	Population or data	Published result used	Use during development	Use during evaluation	Independence status
Hubacher et al. (2006) (23)	Randomized copper intrauterine-device insertion study with literature context	Clinical context for copper-device effects	Initial source-registry context; the 1.2-day bleeding shift is an investigator calibration constant	No direct quantitative target	Development context
Faundes et al. (1980); Malmqvist et al. (1974) (24,25)	Copper-device cycle-hormone study and measured menstrual blood loss	Ovarian cycling persists and menstrual bleeding increases	Later corroboration of the copper-device profile	Internal ovulation and bleeding-direction checks	Later corroboration
Dawood (2006) (26)	Clinical review of primary dysmenorrhea	Predominantly ovulatory, prostaglandin-mediated pain phenotype	Small bleeding-duration modifier	Internal preserved-ovulation and bleeding checks	Development source reused internally

75 1.3 Workflow and data hierarchy

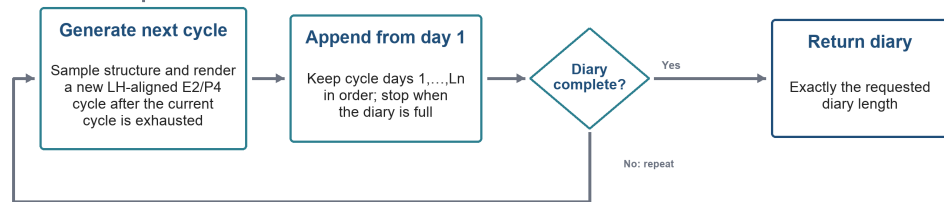
HORMONE-CYCLE diary-generation workflow

Each cycle: timing → LH-aligned serum envelope → long-follicular E2 branch → PCHIP + bridged serial noise.

INITIALIZATION, WAVEFORM, AND FIRST CYCLE



REPEATED COMPLETE-CYCLE GENERATION



Appendix Figure A1. HORMONE-CYCLE diary-generation workflow. Each generated cycle samples structure and bleeding, maps the LH-aligned daily serum envelope to the realized ovulation and luteal interval, applies the long-follicular E2 branch when applicable, and adds endpoint-bridged serial noise after PCHIP interpolation. Cycle 1 is generated in full and diary day 1 is selected uniformly from its realized cycle days. Output proceeds forward through the remainder of cycle 1 and successive complete cycles until the requested diary length is reached; there is no wrapping.

1.3.1 Inputs and outputs

The public simulation call accepts age in years, diary length in days, a random seed, and optional reproductive or medical modifiers. Input validation rejects nonfinite ages, nonpositive diary lengths, and incompatible modifier combinations. The default start mode generates cycle 1 in full, selects one of its realized cycle days with equal probability, and assigns that day to diary day 1. The optional cycle-day-1 mode retains the original observation boundary. After the initial selection, records are retained in chronological order without wrapping. Another cycle is generated only when the retained diary remains shorter than the requested length. Records from the final generated cycle are retained only until the exact requested number of days is reached. Each daily row contains the calendar-day index, cycle index, day within cycle, realized cycle length, estradiol in picograms per milliliter, progesterone in nanograms per milliliter, ovulation, and bleeding. Cycle summaries report phase lengths, ovulation day, ovulatory status, bleeding days, reproductive stage, and active modifiers.

1.4 Age-specific patient-level parameters

Age selects one of eight calibration bands. Li et al. reported 165,668 cycles from 12,608 participants and defined an irregular participant using a mean adjacent-cycle difference of at least 7 days. Their table footnote omits ‘absolute,’ but their methods define adjacent-cycle differences as absolute values; version 0.3.0 follows that convention. The model uses their age-specific pooled within-person SD, participant irregularity prevalence, and short- and long-cycle tails; age-band means are derived from the published adjusted contrasts anchored to the reported overall mean. (5) Participant irregularity is therefore evaluated with Equation (1) rather than as the probability that any single adjacent pair crosses the threshold.

$$I_i = \mathbf{1} \left\{ \frac{1}{n_i - 1} \sum_{j=2}^{n_i} |L_{ij} - L_{i,j-1}| \geq 7 \right\} \quad (1)$$

where I_i is the participant-level irregularity indicator, equal to one when the mean absolute adjacent-cycle difference is at least seven days and zero otherwise; i is the participant index; n_i is the number of retained cycles for participant i ; j is the ordered cycle index; L_{ij} is the length in days of cycle j for participant i ; $L_{i,j-1}$ is the preceding cycle length for the same participant.

Each participant is assigned once to an age-specific low- or high-variability component. A deterministic shifted-lognormal surrogate supplied initial component probabilities and standard deviations against four AWHs outcomes per age band: pooled within-person SD, participant irregularity prevalence, and short- and long-cycle tails, using 11 cycles per participant to match the source cohort's median follow-up. Production values were then refined and accepted against the complete simulator, which additionally represents between-person means, rounding and truncation, random observation boundaries, and variable realized follow-up. Given the resolved cycle-specific latent mean and SD, Equations **Error! Reference source not found.** and (3) parameterize and sample a shifted lognormal cycle length. The 17-day shift supplies realistic right skew while the 18–120-day limits remain broad software safeguards. The ≥ 50 -year band also includes an occasional long-cycle episode fitted to its joint targets.

$$\tau_{ij}^2 = \ln \left[1 + \left(\frac{s_{ij}}{m_{ij} - a} \right)^2 \right], \lambda_{ij} = \ln(m_{ij} - a) - \frac{\tau_{ij}^2}{2}, a = 17 \quad (2)$$

where τ_{ij} is the natural-log-scale standard deviation for the shifted lognormal draw; λ_{ij} is the natural-log-scale mean for the shifted lognormal draw; s_{ij} is the resolved latent cycle-length standard deviation after participant component and cycle-specific modifiers; m_{ij} is the resolved latent cycle-length mean after participant and cycle-specific modifiers and compensation for expected explicit shifts; a is the fixed 17-day shift; i is the participant index; j is the cycle index.

$$L_{ij} = \min(120, \max[18, \text{round}\{a + \exp(\lambda_{ij} + \tau_{ij}Z_{ij}) + E_{ij}\}]), Z_{ij} \sim \mathcal{N}(0,1) \quad (3)$$

where L_{ij} is the realized integer cycle length in days; Z_{ij} is an independent standard-normal cycle draw; E_{ij} is the explicit long-cycle episode extension, zero outside the age-50-or-older episode mechanism; a is the 17-day shift defined in Equation 2; λ_{ij} is the log-scale mean defined in Equation 2; τ_{ij} is the log-scale standard deviation defined in Equation 2; i is the participant index; j is the cycle index; $\mathcal{N}$ is the normal distribution.

Stable person-level estradiol and progesterone amplitude multipliers follow lognormal distributions with coefficients of variation 0.18 and 0.22. The lognormal parameters are chosen so the multiplier has mean one.

$$\sigma_{\log} = \sqrt{\ln(1+c^2)} \quad (4)$$

where $\sigma_{\log}$ is the standard deviation on the natural-log scale; c is the requested coefficient of variation of the positive hormone-amplitude multiplier.

$$\mu_{\log} = -\frac{\sigma_{\log}^2}{2} = -\frac{1}{2} \ln(1+c^2) \quad (5)$$

where $\mu_{\log}$ is the mean on the natural-log scale; $\sigma_{\log}$ is the standard deviation on the natural-log scale; c is the requested coefficient of variation of the positive hormone-amplitude multiplier.

These parameter choices give the lognormal multiplier an arithmetic mean of one.

Appendix Table A2. Age-specific baseline targets. Cycle-length and irregularity values are from Li et al. (2023) (5). Ovulation probabilities are investigator-selected calibration constants implemented in HORMONE-CYCLE. The two rows below age 20 share one Li target band and differ only in the implemented ovulation probability.

Age band (years)	Mean cycle length (days)	Participants irregular: mean absolute adjacent-cycle difference ≥ 7 days	Baseline ovulation probability
<15	29.82	20.4%	0.62
15–19	29.82	20.4%	0.90
20–24	29.62	20.9%	0.97
25–29	29.31	15.9%	0.97
30–34	28.75	11.6%	0.97
35–39	28.19	10.6%	0.97
40–44	27.70	12.8%	0.95
45–49	27.86	28.2%	0.88
≥ 50	30.21	60.2%	0.70

1.5 Cycle length, phase timing, and bleeding

For each cycle, ovulation is drawn from a Bernoulli distribution using the resolved person-level probability. Natural cycle length is drawn from the shifted lognormal process in Equations **Error! Reference source not found.** and (3), rather than the former symmetric Gaussian. Ovulatory cycles use a luteal distribution centered on 12.4 days; the follicular interval absorbs most remaining cycle-length variation. Natural Cycles data reported a 29.3-day mean cycle, 16.9-day mean follicular phase, and 12.4-day mean luteal phase across 612,613 ovulatory cycles from 124,648 users. (6) The implemented latent luteal standard deviation is 3.0 days before truncation and rounding; the validation target is the published 2.4-day realized SD.

Bleeding duration is sampled around 4.0 days with a standard deviation of 1.5 days and is bounded to 0–12 days. Bull et al. reported the same 4.0 ± 1.5 -day distribution in Table 1; both its mean and standard deviation are direct validation targets. The International Federation of Gynecology and Obstetrics terminology provides the clinical interpretation framework for bleeding patterns. (12) The cycle-length bounds and dispersion values are transparent calibration choices.

1.6 Daily hormone trajectories

Ovulatory trajectories in version 0.3.0 use the complete daily serum estradiol and progesterone medians measured in 20 healthy volunteers and synchronized to the serum LH peak. (7) The simulator places ovulation 0.75 day after that event. For nonpositive LH-relative days, published one-day spacing is retained; positive offsets are multiplied by the realized post-LH interval divided by 14, so the published +14-day tail reaches the final cycle day. Progesterone uses the complete –15 to +14 series, producing a postovulatory rise, a broad midluteal summit, and late-luteal withdrawal. Estradiol uses the –1 to +14 segment after an ordinary follicular ramp and retains the published secondary luteal elevation. PCHIP interpolation, defined in Equations 6 and 7, connects these daily values without spline overshoot; Equation 8 defines the LH-relative event mapping.

$$t = \frac{x - x_k}{h_k}, h_{00} = 2t^3 - 3t^2 + 1, h_{10} = t^3 - 2t^2 + t, h_{01} = -2t^3 + 3t^2, h_{11} = t^3 - t^2 \quad (6)$$

where t is the normalized position within one interpolation interval; x is the requested cycle-day position; x_k is the position of control point k ; h_k is the interval width $x_{k+1} - x_k$; k is the control-point interval index; h_{00} is the left-value cubic Hermite basis function; h_{10} is the left-slope cubic Hermite basis function; h_{01} is the right-value cubic Hermite basis function; h_{11} is the right-slope cubic Hermite basis function.

$$y(x) = h_{00}y_k + h_{10}h_k m_k + h_{01}y_{k+1} + h_{11}h_k m_{k+1}, x_k \leq x \leq x_{k+1} \quad (7)$$

where $y(x)$ is the interpolated hormone concentration at position x ; x is the requested cycle-day position; x_k is the position of control point k ; x_{k+1} is the position of the next control point; y_k is the hormone concentration at control point k ; y_{k+1} is the hormone concentration at the next control point; h_k is the interval width $x_{k+1} - x_k$; m_k is the shape-preserving derivative at control point k ; m_{k+1} is the shape-preserving derivative at the next control point; h_{00} is the left-value basis function from Equation 6; h_{10} is the left-slope basis function from Equation 6; h_{01} is the right-value basis function from Equation 6; h_{11} is the right-slope basis function from Equation 6; k is the control-point interval index.

$$x(r) = \begin{cases} o - \delta + r, & r \leq 0, \\ o - \delta + r \frac{L - (o - \delta)}{14}, & r > 0, \end{cases} \delta = 0.75 \quad (8)$$

where $x(r)$ is the simulated cycle-day position assigned to published LH-relative offset r ; r is the integer day relative to the synchronized serum LH peak in the Stricker daily series; o is the simulated ovulation day; δ is the fixed 0.75-day interval from the serum LH peak to the simulated ovulation event; L is the realized cycle length in days; 14 is the final positive offset in the published daily reference series.

Appendix Table A3. Complete daily ovulatory serum-hormone reference series used by HORMONE-CYCLE version 0.3.0, derived from Stricker et al. (2006) Table 1B and Figure 1. (7) Day 0 is the synchronized serum LH peak; the simulator ovulation marker is 0.75 day later. Estradiol was converted from picomoles per liter to picograms per milliliter and progesterone from nanomoles per liter to nanograms per milliliter.

Day relative to serum LH peak	Estradiol median (pg/mL)	Progesterone median (ng/mL)
-15	37.99	0.40
-14	40.59	0.40
-13	37.49	0.30
-12	34.99	0.20
-11	35.49	0.20
-10	39.54	0.20
-9	41.99	0.20
-8	44.34	0.15

Day relative to serum LH peak	Estradiol median (pg/mL)	Progesterone median (ng/mL)
-7	53.44	0.20
-6	58.58	0.10
-5	71.43	0.15
-4	98.92	0.20
-3	132.31	0.10
-2	177.35	0.20
-1	255.88	0.30
+0	182.80	0.80
+1	85.23	1.55
+2	70.98	4.00
+3	87.98	6.45
+4	109.92	9.70
+5	122.77	10.70
+6	132.56	11.10
+7	150.31	13.35
+8	133.81	11.65
+9	137.16	10.20
+10	134.96	10.30
+11	92.73	4.55
+12	85.68	4.10
+13	46.34	1.65
+14	41.19	1.30

Long follicular phases are not created by horizontally stretching the ordinary estradiol curve. When the realized follicular interval is at least 24 days, the model keeps a 14-day terminal maturation segment aligned to ovulation and assigns extra time to a low early interval. A deterministic patient-and-cycle hash selects delayed dominant-follicle emergence by default or a failed-wave/replacement geometry in 25% of such cycles. Harlow et al. reported five heterogeneous urinary-oestrogen patterns in 28 long follicular phases, with delayed emergence most common; the implementation represents only two of those patterns. (9) Mumford et al. independently found that in cycles longer than 35 days, E2 and LH peaks occurred about three days later than in normal-length cycles, supporting delayed timing rather than proportional stretching. (10) The 25% share and failed-wave amplitude are investigator-selected heterogeneity settings, not published prevalence estimates. Equation 9 defines the start of the terminal maturation interval.

$$b = \max(2, o - 14) \text{ for } o \geq 24 \quad (9)$$

where b is the first cycle day of the terminal follicular estradiol-maturation interval; o is the follicular length and simulated ovulation day; 14 is the fixed terminal maturation duration in days; 24 is the follicular-length threshold in days for the long-follicular geometry.

Anovulatory cycles use four lower-amplitude control points. Person-level hormone scales are multiplied by cycle-level lognormal scales with coefficients of variation 0.08 for estradiol and 0.10 for progesterone. Day-to-day continuity is produced by a stationary first-order autoregressive process with coefficient 0.92. Each realized noise path is linearly bridged to zero at both cycle endpoints so stochastic noise cannot recreate a vertical reset.

$$\begin{aligned} \tilde{\epsilon}_d &= \phi \tilde{\epsilon}_{d-1} + \eta_d, \eta_d \sim \mathcal{N}(0, s^2(1 - \phi^2)), \phi = 0.92; \epsilon_d \\ &= \tilde{\epsilon}_d - \left[\tilde{\epsilon}_1 + \frac{d-1}{L-1} (\tilde{\epsilon}_L - \tilde{\epsilon}_1) \right] \end{aligned} \quad (10)$$

where $\tilde{\epsilon}_d$ is the unbridged fractional serial-noise state on day d ; ϵ_d is the endpoint-bridged fractional serial-noise value applied on day d ; d is the cycle-day index; L is the realized cycle length in days; ϕ is the first-order autoregressive coefficient fixed at 0.92; η_d is the independent Gaussian innovation on day d ; s is the stationary standard deviation of the hormone-specific fractional noise process; $\mathcal{N}$ is the normal distribution.

The stationary innovation scale preserves the requested marginal variance, and the linear bridge forces the fractional noise contribution to zero on the first and final cycle days. Each interpolated hormone value is multiplied by $1 + \epsilon_d$ before the estradiol floor of 5 picograms per milliliter or progesterone floor of 0.05 nanograms per milliliter is applied.

1.7 Reproductive-stage and medical modifiers

1.7.1 Polycystic ovary syndrome

The polycystic-ovary profile lengthens cycles, increases cycle-to-cycle variation, reduces ovulation probability, and attenuates luteal progesterone. Mortimer et al. (2026) and Doi et al. supplied the initial direction-setting evidence. (13,14) Later age-stratified cycle data and intensive ultrasound and hormone sampling corroborated the implemented profile. (13,15)

1.7.2 Early postmenarche

The early-postmenarche profile lengthens cycles, increases variability, reduces ovulation, and shortens or attenuates luteal progesterone patterns. Venturoli et al. supplied the initial direction-setting evidence. (16) An adolescent multicenter study and a two-year daily hormone study later corroborated these directions. (17,18)

1.7.3 Perimenopause

The perimenopause profile increases cycle variability, lowers ovulation probability, attenuates progesterone, and increases hormone variability in accordance with menopause-transition evidence. (19)

1.7.4 Combined oral contraceptives

Cyclic and continuous combined oral-contraceptive profiles use fixed 28-day regimen cycles with suppressed endogenous-equivalent hormone curves. The cyclic profile generates scheduled withdrawal bleeding. The continuous profile generates amenorrhea or breakthrough bleeding. These behaviors follow the evidence summarized by Edelman et al. (20)

1.7.5 Intrauterine devices

The levonorgestrel-releasing intrauterine-device profile preserves ovulation in most cycles and reduces bleeding, with amenorrhea in a minority of cycles. Long-term hormone and ultrasound studies support these directions. (21,22) The copper intrauterine-device profile preserves ovarian cycling and increases bleeding duration. Hubacher et al. supplied contextual evidence in the initial source registry. (23) Hormone and measured-blood-loss studies later corroborated that profile. (24,25)

1.7.6 Primary dysmenorrhea

The primary-dysmenorrhea profile preserves ovulation and applies a small bleeding-duration shift. This implementation reflects the predominantly prostaglandin-mediated pain phenotype described in the clinical literature. (26)

236 **Appendix Table A4. Core HORMONE-CYCLE parameter catalogue. The implemented rules were transcribed from src/hormone_cycler/hormone_constants.py,**
 237 **model.py, and types.py. The final column distinguishes direct or derived published quantities, calibrated constants, investigator choices, and software rules. Published**
 238 **timing and hormone anchors refer to Li et al. (2023), Bull et al. (2019), and Stricker et al. (2006) (5–7). These exact values describe version 0.3.0.**

Parameter	Hierarchy	Implemented rule or value	Bounds, rounding, or transformation	Evidence classification
Software/runtime	Package	HORMONE-CYCLE version 0.3.0; Python ≥ 3.11	Source archive and lock file	Open-source implementation
Random-number streams	Participant and cycle	SHA-256 domain-separated Python random.Random streams for participant profile, cycle generation, and diary-start selection	Exact reproducibility for fixed inputs	Implementation choice preventing observation-boundary selection from perturbing latent profile or cycle draws
Age and stage	Participant	Age and resolved reproductive stage remain fixed for the requested diary	Age 12–<56 in calibration bands; public input is numeric	Study design and Li et al.
Participant mean cycle length	Participant	Truncated Gaussian around the age-band target after compensation for expected anovulatory and ≥ 50 long-cycle shifts	Between-person standard deviation 0.8 day (<20), 0.6 day (20–44), or 0.8 day (≥ 45); range 20–90 days	Derived and calibrated from Li et al.
Within-participant cycle standard deviation	Participant	Age-specific Bernoulli assignment to a low- or high-variability participant component; component standard deviation is stable within person	Deterministic shifted-lognormal surrogate initialization followed by full-simulator refinement against AWHS SD, irregularity, and short/long tails	Derived and calibrated from Li et al.
Natural-cycle length	Cycle	17-day-shifted lognormal draw around the resolved person/cycle mean and variability; rounded to nearest day	18–120 days; age ≥ 50 adds a 25.4912-day episode with probability 0.109 after compensating the latent mean	Bull et al. plus physiologic calibration bounds
Ovulation	Cycle	Bernoulli draw using the participant probability in Table A2	Probability constrained to 0–0.99	Investigator calibration constrained by age/stage evidence
Anovulatory cycle shift	Cycle	+1.5 days in reproductive stage; +2.5 early postmenarche; perimenopause +2.5 with probability 0.65 and –1.25 otherwise	Cycle standard deviation $\times 1.15$	Investigator calibration constants
Luteal duration	Cycle	Truncated Gaussian centered on 12.4 days with latent standard deviation 3.0 days; rounded to nearest day	Standard deviation 1.7 days; lower bound 9; upper bound $\min(17, \text{cycle length} - 8)$	Bull et al. plus calibration bounds

Parameter	Hierarchy	Implemented rule or value	Bounds, rounding, or transformation	Evidence classification
Follicular duration and ovulation day	Cycle	Cycle length minus luteal duration; ovulation day equals follicular duration	Follicular lower bound 7 days	Algorithmic consequence of the stable-luteal design
Bleeding duration	Cycle	Truncated Gaussian centered on 4.0 days with standard deviation 1.5 days; rounded to nearest day	0–12 days	Direct Bull et al. Table 1 target; Fraser terminology
Participant hormone amplitudes	Participant	Mean-one lognormal multipliers	Coefficient of variation 0.18 for estradiol and 0.22 for progesterone	Calibration to Stricker-derived control points
Cycle hormone amplitudes	Cycle	Mean-one lognormal multipliers applied to participant scales	Coefficient of variation 0.08 for estradiol and 0.10 for progesterone	Investigator calibration constants
Ovulatory control-point positions	Cycle/day	Daily Stricker medians indexed to the LH peak; simulated ovulation is 0.75 day later. Nonpositive offsets retain one-day spacing; positive offsets are scaled to the realized post-LH interval	The published +14-day tail reaches the final simulated cycle day without moving the progesterone rise before ovulation	Stricker daily series plus event-alignment rule
Ovulatory hormone anchors	Cycle/day	All 30 daily E2/P4 values in Table A3; ordinary-cycle E2 uses the −1 through +14 segment after a follicular ramp, and P4 uses −15 through +14	Converted from pmol/L and nmol/L; rounded to two decimals for reporting	Direct Stricker medians after unit conversion
Long-follicular estradiol geometry	Cycle/day	For follicular phases ≥ 24 days, excess time precedes a fixed 14-day terminal maturation interval; deterministic profile hashing selects delayed dominant-follicle emergence or a failed-wave/replacement geometry	Failed-wave share 0.25 and failed-wave peak $0.52 \times$ ordinary preovulatory target	Harlow qualitative morphology; Mumford independent timing context; share and amplitude are investigator-selected sensitivity settings
Anovulatory estradiol anchors	Cycle/day	38, 86, 74, and 44 picograms per milliliter at day 1, $0.55 \times$ cycle length, cycle end −3, and cycle end	Smooth interpolation between points	Investigator calibration constrained by anovulatory endocrine literature
Anovulatory progesterone anchors	Cycle/day	0.35, 0.55, 0.75, and 0.40 nanograms per milliliter at the same four positions	Smooth interpolation between points	Investigator calibration constrained by anovulatory endocrine literature
Daily interpolation	Day	Shape-preserving piecewise cubic Hermite interpolation (PCHIP) using the basis and piecewise interpolation defined below	Applied separately to estradiol and progesterone without spline overshoot	Algorithmic choice

Parameter	Hierarchy	Implemented rule or value	Bounds, rounding, or transformation	Evidence classification
Daily serial noise	Day	Stationary first-order autoregressive state with coefficient 0.92; realized paths linearly bridged to zero at both cycle endpoints	Estradiol stationary standard deviation equals participant noise scale; progesterone uses $0.9 \times$ that scale	Investigator smoothing/calibration choice
Hormone reporting	Day	Values rounded to two decimal places	Estradiol floor 5 picograms per milliliter; progesterone floor 0.05 nanograms per milliliter	Reporting and physiologic floor choices
Anovulatory spotting	Cycle/day	Probability 0.25 in early-postmenarche or perimenopause anovulatory cycles; begins at $0.65 \times$ cycle length	Duration 2 days	Investigator stress-test setting
Modifier combinations	Participant	Polycystic ovary syndrome, stage, intrauterine device, and dysmenorrhea adjustments apply sequentially; oral-contraceptive settings override timing and hormone profiles	Copper and levonorgestrel intrauterine devices are mutually exclusive; oral contraceptives cannot coexist with either device	Software validation rules
Diary start and completion	Run	Generate cycle 1 in full; select its first observed day uniformly; continue forward through subsequent cycles	Optional cycle-day-1 start; no wrapping; truncate only the final retained cycle at the requested diary length	Implementation rule

Appendix Table A5. Implemented modifier values and their evidence sources. The numerical multipliers are calibration constants unless the row states a direct published quantity. The cited papers support the direction or clinical context of each profile.

Modifier	Cycle-length effect	Cycle variability	Ovulation effect	Bleeding and hormone effects	Published evidence
Polycystic ovary syndrome	Mean $\times 1.30$ (<25 years), $\times 1.22$ (25–34), $\times 1.15$ (≥ 35)	Variability $\times 1.55$	Probability $\times 0.48$	Progesterone amplitude $\times 0.58$; bleeding +0.4 day	Mortimer et al.; Doi et al.; Jarrett et al. (13,13–15)
Early postmenarche	Mean +2.5 days	Variability $\times 1.25$	Maximum probability 0.55	Estradiol $\times 0.92$; progesterone $\times 0.72$; bleeding +0.5 day	Venturoli et al.; World Health Organization study; Zhang et al. (16–18)
Perimenopause	Mean unchanged	Variability $\times 1.35$	Probability $\times 0.78$	Progesterone $\times 0.82$; bleeding +0.6 day	Santoro and Randolph (19)
Copper intrauterine device	Cycle length unchanged	Cycle variability unchanged	Preserved	Bleeding +1.2 days; bleeding standard deviation +0.25 day	Hubacher et al.; Faundes et al.; Malmqvist et al. (23–25)
Levonorgestrel-releasing intrauterine device	Cycle length unchanged	Cycle variability unchanged	Maximum probability 0.82	Bleeding –2.2 days with 0.8-day floor; amenorrhea probability 0.17	Xiao et al.; Barbosa et al. (21,22)
Cyclic combined oral contraceptive	Fixed 28 days	Standard deviation 0.25 day	Suppressed	Mean scheduled withdrawal bleeding 4.0 days; suppressed endogenous-equivalent curves	Edelman et al. (20)
Continuous combined oral contraceptive	Fixed 28 days	Standard deviation 0.20 day	Suppressed	Amenorrhea probability 0.55; breakthrough bleeding modeled	Edelman et al. (20)

Modifier	Cycle-length effect	Cycle variability	Ovulation effect	Bleeding and hormone effects	Published evidence
Primary dysmenorrhea	Cycle length unchanged	Cycle variability unchanged	Preserved	Bleeding +0.5 day; bleeding standard deviation +0.15 day	Dawood (26)

1.8 Reproducibility and implementation limits

Fixed inputs and a fixed seed produce identical diaries. SHA-256 domain separation creates independent profile, cycle, and observation-start streams, so changing the diary start rule does not perturb the latent person or generated cycles. For computational efficiency, non-audit participants in the large paper simulation use a compact renderer that consumes the same random draws as the full renderer, retains cycle timing, ovulation, midluteal progesterone and inadequate-luteal-phase status, and omits unused daily concentration objects. Validation and the retained 1% audit sample use full daily hormone curves; automated equivalence tests compare both paths, including partial first and final cycles. Reproducibility records should preserve the repository commit, configuration, seed, requested diary length, age and modifier inputs, and validation-report hash. The implementation omits pregnancy, postpartum physiology, medication changes, secular drift in an individual's latent mean or variability component, missing diary entries, bleeding intensity, laboratory-assay error, and mechanistic endocrine feedback. Modifier combinations are coarse profile transformations for synthetic stress testing; causal treatment effects and individual clinical prediction remain outside scope.

Chapter 2. Calibration and validation status of HORMONE-CYCLE

2.1 Evaluation terminology, source separation, and design

This chapter distinguishes calibration-target reproduction, held-out aggregate cross-checks, waveform construction checks, visual checks, and modifier stress tests. AWHs and Bull informed cycle calibration, and the complete Stricker daily series constructs the ovulatory waveform. Cunningham was withheld from cycle fitting, while Anckaert was withheld from hormone-waveform construction; both are independent at the aggregate-source level rather than participant-level validation. Harlow informed long-follicular E2 geometry, and Mumford supplied independent long-cycle timing context.

The quality-control design follows established simulation-study reporting principles. (3,4) The baseline command was equivalent to “hormone_cycler validate --patients 10000 --days 365 --seed 7 --start-mode random.” The validation workflow restricted this baseline cohort to ages 18–54.9 to match AWHs adult eligibility. Secondary modifier scenarios used 1,200 age-matched participants and paired unmodified cohorts generated from the same seed: ages 18–44.9 for PCOS, contraception, IUD, and dysmenorrhea; ages 45–54.9 for perimenopause; and ages 13–17.9 for peri-menarche. Acceptance rules were encoded in src/hormone_cycler/hormone_constants.py before the reported execution. The run evaluated 67 calibration and waveform metrics, 14 held-out Cunningham metrics, and eight modifier scenarios. The primary gate required every calibration, waveform, and held-out metric to pass; modifier results were secondary stress tests.

The AWHs age-band comparisons use equal participant weighting and at most the first 11 cycles, matching the source cohort’s median follow-up. The Cunningham cross-check uses the same simulated 12-month-equivalent summaries but retains the source study’s different age bands and estimator. AWHs’s table footnote says mean adjacent-cycle difference without specifying absolute values; because its methods define adjacent-cycle differences as absolute values, that convention was prespecified here. These design choices prevent the previous error of treating pairwise threshold exceedances as the participant-level irregularity estimand.

Before the definitive validation and paper rerun, a registry-wide provenance audit checked the title, PubMed identifier, DOI, source URL, and declared evidence role for all 20 scientific-source records against PubMed. Five legacy identifiers that resolved to unrelated articles and one citation-label mismatch were corrected. The fitted AWHs, Bull, and Stricker quantities and the held-out Cunningham results were unaffected; the corrections prevent secondary context and modifier sources from being misidentified. The machine-readable audit report is archived with the validation artifacts.

2.2 Healthy-cycle calibration targets and acceptance criteria

The AWHs calibration includes five outcomes in each of eight age bands: mean cycle length, pooled within-person SD, participant prevalence with mean absolute adjacent-cycle difference ≥ 7 days, cycles < 24 days, and cycles > 38 days. Practical margins are ± 0.55 day for means; ± 0.35 day for within-person SD below age 50 and ± 1.0 day at age ≥ 50 ; ± 3.5 percentage points for participant irregularity; and ± 3 percentage points for cycle tails below age 50 and ± 5 points at age ≥ 50 . Bull checks add follicular, luteal, and bleeding means plus luteal and bleeding SDs. These margins assess aggregate similarity, not sampling confidence intervals or individual prediction.

2.3 Held-out aggregate validation and critical interpretation

Cunningham et al. reported age-specific 12-month participant means and mean participant-specific cycle-length SDs from a global Flo cohort. (8) The source was not used for fitting. Prespecified practical-equivalence margins were ± 1.5 days for mean cycle length and ± 1.25 days for mean personal SD, widened to ± 2.5 and ± 3.5 days, respectively, at ages 51–55 because the menopause-transition cohorts and estimators differ substantially.

The external summary check passed, but the interpretation is qualified. Agreement was strongest from ages 18–45. At ages 51–55, the simulator’s variability was higher than Flo but close to the much larger AWHs ≥ 50 residual SD. The two published cohorts therefore disagree in the region with sparse, selective, and rapidly changing cycles. The simulator retains an intermediate high-variability behavior rather than fitting the held-out source. This is a model-form uncertainty, not evidence that one cohort is correct.

2.4 Hormone waveform and phase-duration checks

Fourteen hormone-amplitude checks compare seven estradiol and seven progesterone subphases with the independently measured Anckaert reference intervals; broad assay-aware windows account for platform and cohort differences. Eight daily morphology checks evaluate preovulatory E2 width, the secondary luteal E2 peak, progesterone plateau width, P4 rise and peak timing relative to ovulation, consecutive premenstrual withdrawal, terminal-to-peak ratio, and cross-cycle continuity. The validation implementation retains two diaries from each of eight age bands (16 total). The ordinary-cycle envelope is also overlaid directly on mapped Stricker medians, while 53-day delayed-emergence and failed-wave examples test the no-stretch rule. These are aggregate amplitude, construction, and software morphology checks—not validation of joint person-level endocrine trajectories or intraday progesterone pulses.

314 **Appendix Table A6. Complete healthy-cycle calibration, held-out aggregate cross-check, and hormone waveform results. The 67 calibration and waveform metrics**
 315 **comprise 40 AWHs age-band metrics, five Bull phase/bleeding metrics, 14 independent Anckaert amplitude/order checks, and eight Stricker-derived daily morphology**
 316 **checks. Calibration and construction sources are (5–7) Fourteen Cunningham metrics are held out from cycle fitting. (8) Anckaert was held out from waveform**
 317 **construction. Cycle and phase metrics use 10,000 adults aged 18–54.9, 365 days per participant, seed 7, randomized starting phase, equal participant weighting, and at**
 318 **most 11 cycles per participant. Hormone metrics use 16 retained diaries balanced across eight age bands. Every row met its prespecified acceptance band. This qualified**
 319 **pass does not constitute individual-level external validation. Durations are in days; estradiol is in picograms per milliliter; progesterone is in nanograms per milliliter.**

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Mean cycle length, age <20	29.85	29.82	29.27 to 30.37	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Pooled within-person cycle-length SD, age <20	5.35	5.33	4.98 to 5.68	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Participants with mean absolute adjacent difference ≥7 days, age <20	20.4%	20.4%	16.9% to 23.9%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles shorter than 24 days, age <20	6.5%	7.7%	4.7% to 10.7%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles longer than 38 days, age <20	6.4%	8.3%	5.3% to 11.3%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Mean cycle length, age 20–24	29.55	29.62	29.07 to 30.17	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Pooled within-person cycle-length SD, age 20–24	5.15	5.07	4.72 to 5.42	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Participants with mean absolute adjacent difference ≥ 7 days, age 20–24	23.0%	20.9%	17.4% to 24.4%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles shorter than 24 days, age 20–24	7.5%	7.2%	4.2% to 10.2%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles longer than 38 days, age 20–24	6.2%	6.9%	3.9% to 9.9%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Mean cycle length, age 25–29	29.34	29.31	28.76 to 29.86	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Pooled within-person cycle-length SD, age 25–29	4.71	4.70	4.35 to 5.05	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Participants with mean absolute adjacent difference ≥ 7 days, age 25–29	17.0%	15.9%	12.4% to 19.4%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Cycles shorter than 24 days, age 25–29	6.9%	6.1%	3.1% to 9.1%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles longer than 38 days, age 25–29	4.8%	6.2%	3.2% to 9.2%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Mean cycle length, age 30–34	28.75	28.75	28.20 to 29.30	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Pooled within-person cycle-length SD, age 30–34	4.33	4.28	3.93 to 4.63	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Participants with mean absolute adjacent difference ≥ 7 days, age 30–34	12.2%	11.6%	8.1% to 15.1%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles shorter than 24 days, age 30–34	8.0%	6.7%	3.7% to 9.7%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles longer than 38 days, age 30–34	3.4%	4.5%	1.5% to 7.5%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Mean cycle length, age 35–39	28.22	28.19	27.64 to 28.74	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Pooled within-person cycle-length SD, age 35–39	3.87	3.79	3.44 to 4.14	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Participants with mean absolute adjacent difference ≥ 7 days, age 35–39	10.6%	10.6%	7.1% to 14.1%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles shorter than 24 days, age 35–39	8.1%	8.1%	5.1% to 11.1%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles longer than 38 days, age 35–39	2.4%	2.9%	0.0% to 5.9%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Mean cycle length, age 40–44	27.69	27.70	27.15 to 28.25	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Pooled within-person cycle-length SD, age 40–44	4.03	3.99	3.64 to 4.34	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Participants with mean absolute adjacent difference ≥ 7 days, age 40–44	13.4%	12.8%	9.3% to 16.3%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles shorter than 24 days, age 40–44	11.2%	11.9%	8.9% to 14.9%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles longer than 38 days, age 40–44	2.6%	2.5%	0.0% to 5.5%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Mean cycle length, age 45–49	27.96	27.86	27.31 to 28.41	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Pooled within-person cycle-length SD, age 45–49	5.67	5.42	5.07 to 5.77	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Participants with mean absolute adjacent difference ≥ 7 days, age 45–49	28.8%	28.2%	24.7% to 31.7%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles shorter than 24 days, age 45–49	17.5%	16.0%	13.0% to 18.9%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Cycles longer than 38 days, age 45–49	4.9%	4.2%	1.2% to 7.2%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Mean cycle length, age ≥ 50	30.06	30.21	29.66 to 30.76	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Pooled within-person cycle-length SD, age ≥ 50	11.15	11.19	10.19 to 12.19	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Participants with mean absolute adjacent difference ≥ 7 days, age ≥ 50	61.4%	60.2%	56.7% to 63.7%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles shorter than 24 days, age ≥ 50	20.3%	18.7%	13.7% to 23.7%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Cycles longer than 38 days, age ≥ 50	14.8%	14.8%	9.8% to 19.8%	Pass	Li et al. (2023), Tables 4–5 and Supplementary Table 2	10,000 adults aged 18–54.9
Mean follicular interval	16.17	16.90	15.90 to 17.90	Pass	Bull et al. (2019), Table 1	10,000 adults aged 18–54.9
Mean luteal interval	12.57	12.40	11.70 to 13.10	Pass	Bull et al. (2019), Table 1	10,000 adults aged 18–54.9
Mean bleeding duration	4.01	4.00	3.60 to 4.40	Pass	Bull et al. (2019), Table 1	10,000 adults aged 18–54.9

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Luteal-interval SD	2.04	2.40	2.00 to 2.80	Pass	Bull et al. (2019), Table 1	10,000 adults aged 18–54.9
Bleeding-duration SD	1.50	1.50	1.25 to 1.75	Pass	Bull et al. (2019), Table 1	10,000 adults aged 18–54.9
Estradiol, early follicular	46.79	34.05	13.62 to 62.99	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHs age band
Progesterone, early follicular	0.38	0.12	0.05 to 0.65	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHs age band
Estradiol, mid follicular	80.97	46.85	18.74 to 86.68	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHs age band
Progesterone, mid follicular	0.19	0.07	0.05 to 0.65	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHs age band
Estradiol, pre ovulatory	181.11	126.40	50.56 to 233.83	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHs age band

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Progesterone, pre ovulatory	0.21	0.06	0.05 to 0.65	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHS age band
Estradiol, ovulation	94.40	222.56	89.02 to 411.73	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHS age band
Progesterone, ovulation	1.45	0.50	0.05 to 2.00	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHS age band
Estradiol, early luteal	93.91	106.24	42.49 to 196.54	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHS age band
Progesterone, early luteal	7.79	7.11	3.20 to 13.15	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHS age band
Estradiol, mid luteal	134.35	137.56	55.03 to 254.50	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHS age band

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Progesterone, mid luteal	11.83	12.33	5.55 to 22.80	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHs age band
Estradiol, late luteal	61.48	107.87	43.15 to 199.56	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHs age band
Progesterone, late luteal	2.77	5.72	2.58 to 10.59	Pass	Anckaert et al. (2021), independent assay-specific subphase reference intervals	16 retained diaries, two per AWHs age band
Estradiol preovulatory peak width at $\geq 80\%$ maximum	3.00	3.00	2.00 to 5.00	Pass	Stricker daily series; prespecified morphology software-check bound	16 retained diaries, two per AWHs age band
Luteal estradiol peak / preovulatory maximum	0.6238	0.5900	0.3500 to 0.8000	Pass	Stricker daily series; prespecified morphology software-check bound	16 retained diaries, two per AWHs age band
Progesterone plateau width at $\geq 75\%$ maximum	6.00	6.00	3.00 to 9.00	Pass	Stricker daily series; prespecified morphology software-check bound	16 retained diaries, two per AWHs age band
Progesterone peak offset from ovulation	6.00	6.00	3.00 to 9.00	Pass	Stricker daily series; prespecified morphology software-check bound	16 retained diaries, two per AWHs age band

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Progesterone ≥ 5 ng/mL rise offset from ovulation	2.00	2.00	1.00 to 4.00	Pass	Stricker daily series; prespecified morphology software-check bound	16 retained diaries, two per AWHs age band
Consecutive progesterone-decline transitions before bleeding	6.00	4.00	3.00 to 8.00	Pass	Stricker daily series; prespecified morphology software-check bound	16 retained diaries, two per AWHs age band
Final-cycle progesterone / cycle maximum	0.0336	0.0400	0.0000 to 0.2000	Pass	Stricker daily series; prespecified morphology software-check bound	16 retained diaries, two per AWHs age band
Progesterone jump across cycle boundary	0.05	0.00	0.00 to 1.00	Pass	Stricker daily series; prespecified morphology software-check bound	16 retained diaries, two per AWHs age band
Held-out mean cycle length, age 18–25	29.66	28.99	27.49 to 30.49	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean participant-specific SD, age 18–25	4.78	4.14	2.89 to 5.39	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean cycle length, age 26–30	29.24	28.76	27.26 to 30.26	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean participant-specific SD, age 26–30	4.19	3.95	2.70 to 5.20	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Held-out mean cycle length, age 31–35	28.66	28.28	26.78 to 29.78	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean participant-specific SD, age 31–35	3.75	3.80	2.55 to 5.05	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean cycle length, age 36–40	28.11	27.68	26.18 to 29.18	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean participant-specific SD, age 36–40	3.00	3.72	2.47 to 4.97	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean cycle length, age 41–45	27.71	27.18	25.68 to 28.68	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean participant-specific SD, age 41–45	3.34	3.92	2.67 to 5.17	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean cycle length, age 46–50	28.46	27.15	25.65 to 28.65	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean participant-specific SD, age 46–50	5.48	4.72	3.47 to 5.97	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant
Held-out mean cycle length, age 51–55	30.00	28.00	25.50 to 30.50	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant

Metric	Simulated	Calibration target	Acceptance range	Result	Source and location	Simulation sample
Held-out mean participant-specific SD, age 51–55	9.78	6.52	3.02 to 10.02	Pass	Cunningham et al. (2024), held-out aggregate cross-check	10,000 adults aged 18–54.9; first 11 cycles per participant

2.5 Age-matched modifier software stress tests

Each modifier was simulated separately with 1,200 participants and compared with an unmodified cohort generated from the same seed and age range. PCOS, oral-contraceptive, intrauterine-device, and dysmenorrhea scenarios used ages 18–44.9; perimenopause used ages 45–54.9; and peri-menarche used ages 13–17.9. The checks assess cycle length, irregularity, ovulation, bleeding, or amenorrhea according to the feature addressed by the cited literature. The papers generally support effect direction or broad ranges; exact numerical margins are investigator-selected regression guards. Accordingly, these are age-matched software stress tests, not held-out clinical validation.

Appendix Table A7. Secondary age-matched modifier stress-test summary. Each scenario uses 1,200 participants and an unmodified cohort generated with the same seed and age range. “Code-defined checks” counts the criteria listed individually in Table A8. Exact margins are investigator-selected regression guards; passing indicates behavior in the declared direction/range, not clinical validity. Supporting sources are attached directly to each scenario: polycystic ovary syndrome is compared with Mortimer et al. (2026) and Jarrett et al. (2020) (13,15). Early postmenarche is compared with the World Health Organization adolescent study (1986) and Zhang et al. (2008) (17,18). Perimenopause is compared with Santoro and Randolph (2011) (19). Combined oral contraceptives are compared with Edelman et al. (2014) (20). The levonorgestrel-releasing intrauterine device is compared with Xiao et al. (1995) and Barbosa et al. (1995) (21,22). The copper intrauterine device is compared with Faundes et al. (1980) and Malmqvist et al. (1974) (24,25). Primary dysmenorrhea is compared with Dawood (2006) (26).

Scenario	Mean cycle (days)	Ovulatory cycles	Mean bleeding (days)	Irregularity	Amenorrhea	Code-defined checks
Polycystic ovary syndrome (ages 18–44.9)	35.77	46.5%	4.40	56.1%	0.2%	3/3 passed
Cyclic combined oral contraceptive (ages 18–44.9)	28.00	0.0%	3.99	0.0%	0.0%	2/2 passed
Continuous combined oral contraceptive (ages 18–44.9)	28.00	0.0%	0.96	0.0%	55.0%	3/3 passed
Levonorgestrel-releasing intrauterine device (ages 18–44.9)	29.00	81.5%	1.68	14.9%	24.0%	2/2 passed
Copper intrauterine device (ages 18–44.9)	28.77	96.2%	5.20	13.3%	0.2%	2/2 passed

Scenario	Mean cycle (days)	Ovulatory cycles	Mean bleeding (days)	Irregularity	Amenorrhea	Code-defined checks
Perimenopause (ages 45–54.9)	29.52	48.0%	5.24	54.5%	0.2%	2/2 passed
Early postmenarche (ages 13–17.9)	33.37	53.9%	5.13	47.6%	0.2%	3/3 passed
Primary dysmenorrhea (ages 18–44.9)	28.82	96.2%	4.51	15.6%	0.5%	2/2 passed

334

335 **Appendix Table A8. Auditable secondary modifier stress-test criteria. Each scenario uses 1,200 participants and a seed- and age-matched unmodified cohort. Acceptance**
336 **bounds were defined in the validation code and are investigator-selected unless the criterion names a direct published quantity. The supporting papers are named in the**
337 **source column; they generally support directions or broad ranges rather than the exact numerical bounds.**

Scenario	Code-defined criterion	Simulated value	Acceptance range	Supporting paper	Result
Polycystic ovary syndrome	Direction-only check: PCOS should shift mean cycle length upward versus its age-matched baseline.	35.77 days	30.75 days to 999.00 days	Mortimer et al. 2026	Pass
Polycystic ovary syndrome	Direction-only check: PCOS should increase cycle irregularity versus its age-matched baseline.	56.1%	21.6% to 100.0%	Mortimer et al. 2026	Pass
Polycystic ovary syndrome	Direction-only check: PCOS should reduce ovulation frequency versus its age-matched baseline.	46.5%	0.0% to 77.9%	Doi et al. 2005	Pass
Cyclic combined oral contraceptive	21/7 combined OCPs should suppress ovulation.	0.0%	0.0% to 2.0%	Edelman et al. 2014	Pass
Cyclic combined oral contraceptive	21/7 regimens should preserve scheduled withdrawal bleeding.	3.99 days	3.00 days to 5.00 days	Edelman et al. 2014	Pass
Continuous combined oral contraceptive	Continuous combined OCPs should suppress ovulation.	0.0%	0.0% to 2.0%	Edelman et al. 2014	Pass
Continuous combined oral contraceptive	Continuous use should reduce bleeding days relative to cyclic use.	0.96 days	0.00 days to 2.50 days	Edelman et al. 2014	Pass

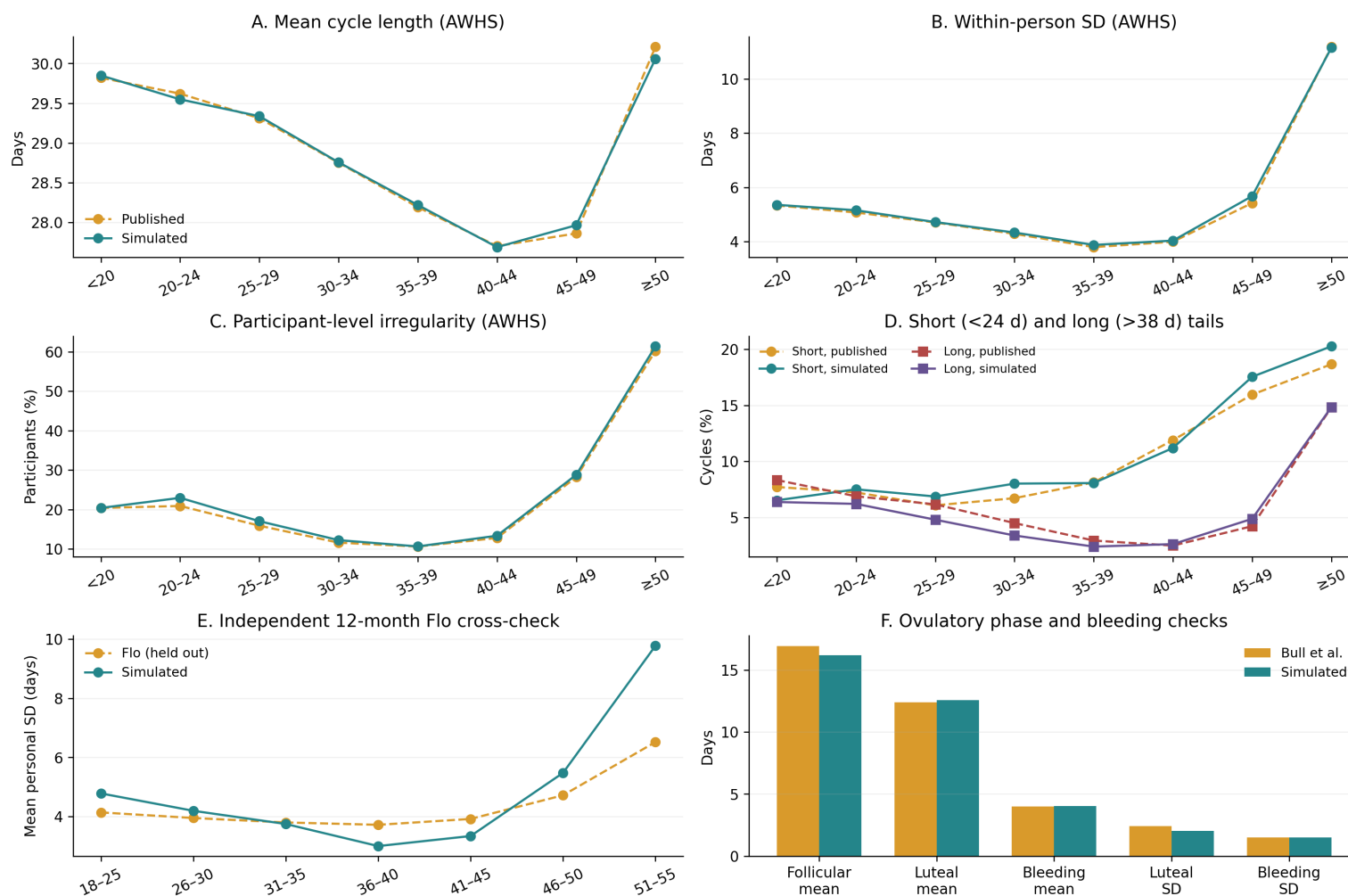
Scenario	Code-defined criterion	Simulated value	Acceptance range	Supporting paper	Result
Continuous combined oral contraceptive	Continuous use should produce substantial amenorrhea/infrequent bleeding.	55.0%	40.0% to 90.0%	Edelman et al. 2014	Pass
Levonorgestrel-releasing intrauterine device	Long-term LNG-IUD users should remain ovulatory in most cycles.	81.5%	70.0% to 90.0%	Xiao et al. 1995	Pass
Levonorgestrel-releasing intrauterine device	Amenorrhea should occur in a minority but not rare fraction of LNG-IUD cycles.	24.0%	10.0% to 25.0%	Xiao et al. 1995	Pass
Copper intrauterine device	Direction-only check: copper IUDs should not materially suppress ovulation versus an age-matched baseline.	96.2%	91.2% to 101.2%	Faundes et al. 1980	Pass
Copper intrauterine device	Direction-only check: copper IUDs should increase bleeding duration versus an age-matched baseline.	5.20 days	4.51 days to 6.01 days	Malmqvist et al. 1974	Pass
Perimenopause	Direction-only check: perimenopause should increase cycle irregularity.	54.5%	25.0% to 100.0%	Santoro and Randolph 2011	Pass
Perimenopause	Direction/range check: perimenopause should reduce ovulation frequency.	48.0%	45.0% to 80.0%	Santoro and Randolph 2011	Pass
Early postmenarche	Direction/range check: early postmenarche cycles should be longer on average.	33.37 days	30.00 days to 40.00 days	WHO Task Force 1986	Pass
Early postmenarche	Direction-only check: early postmenarche cycles should be more irregular.	47.6%	28.0% to 100.0%	WHO Task Force 1986	Pass
Early postmenarche	Direction/range check: adolescents with irregular menses should be frequently anovulatory; this is not a universal perimenarche estimate.	53.9%	30.0% to 60.0%	Venturoli et al. 1986	Pass

Scenario	Code-defined criterion	Simulated value	Acceptance range	Supporting paper	Result
Primary dysmenorrhea	Direction-only check: primary dysmenorrhea should preserve ovulation versus an age-matched baseline.	96.2%	88.2% to 104.2%	Dawood 2006	Pass
Primary dysmenorrhea	Investigator-bounded check: dysmenorrhea should only mildly shift bleeding duration versus an age-matched baseline.	4.51 days	4.21 days to 5.21 days	Dawood 2006	Pass

338

339 2.6 Graphical calibration summary and representative traces

Healthy-cycle simulator calibration and held-out validation



340

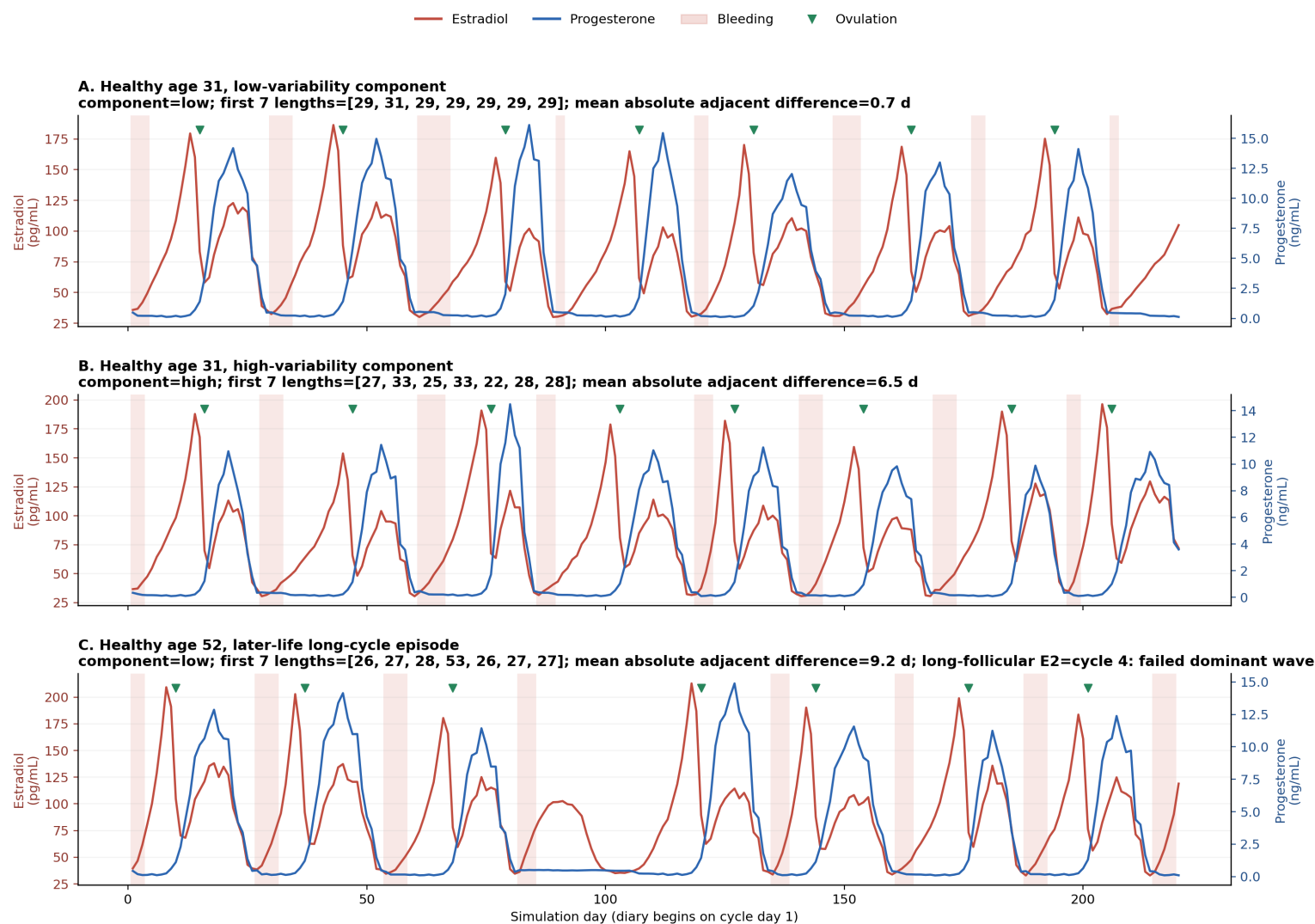
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Appendix Figure A2. Healthy-cycle calibration and held-out aggregate validation. Panels A–D compare simulated age-specific mean cycle length, pooled within-person SD, participant irregularity, and short/long tails with AWHS. Panel F compares follicular, luteal, and bleeding means and SDs with Bull et al. Calibration sources are

343 (5–7) Panel E compares mean participant-specific SD with the held-out Cunningham Flo cohort. (8) Published and simulated lines are aggregate summaries; the older-
344 age discrepancy in panel E is retained as model-form uncertainty.

Healthy-cycle examples after daily hormone-waveform recalibration

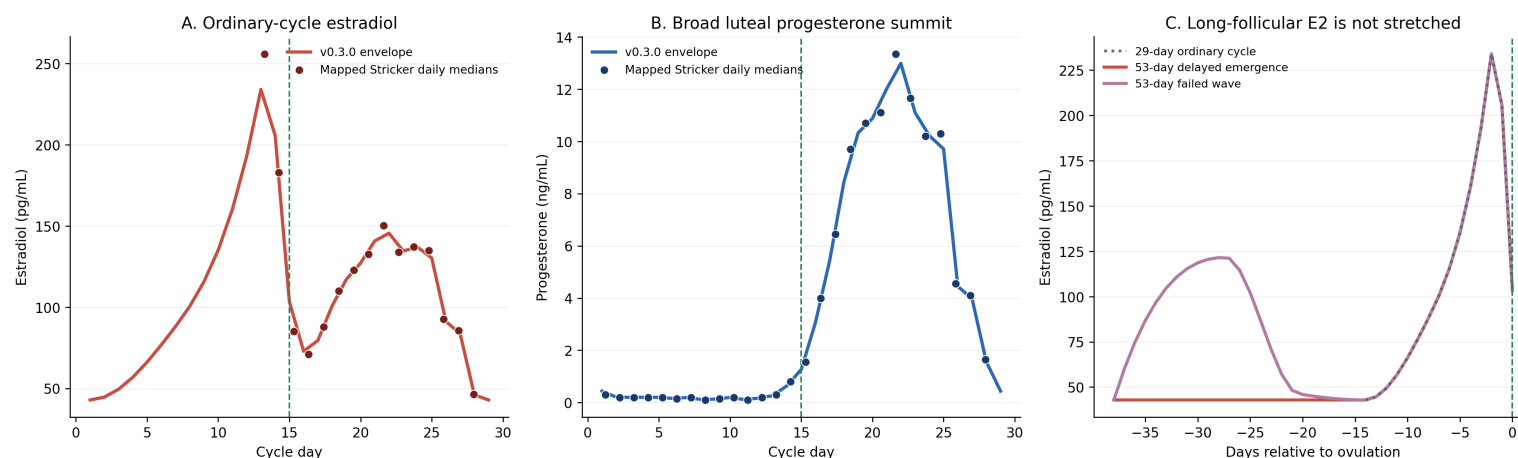


Examples were selected by prespecified display criteria and are illustrative, not validation observations.

Appendix Figure A3. Prespecified illustrative healthy-cycle traces after the version 0.3.0 waveform update. Panel A shows a 31-year-old participant assigned to the low-variability component; panel B shows a 31-year-old participant assigned to the high-variability component; panel C shows a 52-year-old participant with a later-life

long-cycle episode. Estradiol and progesterone use separate axes; red shading denotes bleeding and green triangles denote ovulation. The updated P4 envelope has a postovulatory rise, broad summit, and late-luteal withdrawal. In the long cycle, extra follicular days precede a terminal E2-maturation interval rather than stretching the ordinary template. The displayed seeds were selected by prespecified component and episode criteria; these examples illustrate model behavior and are not independent validation observations.

HORMONE-CYCLE v0.3.0 waveform construction and visual validation



Appendix Figure A4. HORMONE-CYCLE version 0.3.0 waveform construction and visual validation. Panels A and B overlay the ordinary-cycle estradiol and progesterone envelopes on the complete mapped daily Stricker serum medians. (7) The vertical dashed line is the simulated ovulation day, 0.75 day after the synchronized LH peak. Panel C compares an ordinary follicular E2 segment with two Harlow-informed 53-day-cycle geometries. (9) Excess follicular time precedes a fixed 14-day terminal maturation interval; the failed-wave curve is a deterministic heterogeneity option. The 25% failed-wave share is an investigator-selected sensitivity parameter, not an estimated prevalence. The figure checks construction and morphology and is not participant-level clinical validation.

2.7 Validation conclusion and remaining needs

The primary gate is a qualified pass: version 0.3.0 reproduces all prespecified fitted aggregate targets, passes every held-out Cunningham cycle margin and independent Anckaert hormone bound, and passes all daily morphology checks. The P4 envelope is now sustained across several midluteal days rather than concentrated in a single triangular peak; its ≥ 5 -ng/mL rise occurs after ovulation and withdrawal approaches the next menses. Ordinary-cycle E2 retains its recognizable follicular peak and secondary luteal elevation. Long-cycle E2 no longer reflects simple horizontal stretching, but only two of Harlow's five reported long-follicular patterns are implemented, and their mixture is not prevalence-calibrated. The daily population-median envelope also omits intraday hormone pulsatility and does not reproduce individual longitudinal endocrine trajectories. Independent participant-level validation should use held-out daily bleeding, ovulation, and hormone records; align estimands prospectively; and examine joint distributions, serial correlation, between-person heterogeneity, assay effects, and missingness.

Chapter 3. Additional analyses and methods augmenting the main results

This chapter documents how HORMONE-CYCLE and the seizure-diary simulator were combined for the present false-positive study. It then reports the phase algorithms, ratio and regression methods, sensitivity analyses, and supplementary results.

3.1 Reproducibility, provenance, and simulator inputs

The completed primary simulation included 100,000 synthetic participants, 50,000 per cohort, followed for 36 months with master seed 20260505. HORMONE-CYCLE generated menstrual and hormone diaries. (1) The Cyclic Heterogeneous Overdispersed Clustered Open-source L-relationship Adjustable Temporally limited E-diary Simulator (CHOCOLATES) generated seizure diaries. The generators were run independently. HORMONE-CYCLE selected diary day 1 uniformly from the realized days of the first generated menstrual cycle and then continued forward without wrapping. The diaries were merged by calendar-day index. (2) Direct alignment preserved the original day ordering throughout the merge.

The reporting framework follows simulation-study guidance. (3,4) CHOCOLATES and HORMONE-CYCLE were designed to reproduce selected statistical features within and across synthetic patient populations through data-informed hierarchical sampling. Chapter 1 details the HORMONE-CYCLE evidence sources and transformations.

The completed manifest stores the analysis-code fingerprint
 f9101b5895e86df3f86f10e6fda434f62b1e8971b27346b8d20757a054840d73 and configuration fingerprint
 9a1c7f77d1bc8c771dbaf554af89386717f360922f6e13f836c774dd0c7e6c20. These Secure Hash Algorithm
 256-bit fingerprints identify the exact v13 analysis code and configuration used for the completed run.

391 **Appendix Table S1. Simulator inputs, realized features, and evidence roles for the completed study. Literature sources are named in the source/rationale column;**
 392 **investigator-configured values are study assumptions. The named HORMONE-CYCLE, CHOCOLATES, seizure-rhythm, menstrual-cycle, and Herzog sources are**
 393 **cited here (1,2,5–7,27,28). “Simulator-native” means drawn directly by CHOCOLATES. “Multidien” denotes periods longer than 24 hours. “Audit spectrum” is a**
 394 **spectral estimate from retained daily audit data. “Interseizure constraints” are generator limits on simulated intervals between seizures. Hormone concentrations are**
 395 **written in nanograms per milliliter or picograms per milliliter.**

Domain	Parameter	Cohort	Setting / sampling distribution	Level	Source / rationale	Realized target
Seizure	Monthly burden	Both	Simulator-native empirical distribution	Participant	Goldenholz & Westover 2023	Realized seizures/month
Seizure	Overdispersion	Both	Simulator-native	Participant /day	Goldenholz & Westover 2023	Audit daily variance/mean
Seizure	Clustering	Both	Simulator-native	Participant /day	Goldenholz & Westover 2023	Audit next-day clustering proxy
Seizure	Multidien periods and amplitudes	Both	Simulator-native	Participant /day	Goldenholz & Westover 2023; Karoly et al.	Generator period and audit spectrum
Seizure	Interseizure constraints	Both	Simulator-native	Day	Goldenholz & Westover 2023	Not separately stored
Menstrual	Age	Cohort-specific	{'healthy_ovulatory': {'age_range': [18.0, 45.0], 'force_ovulatory_cycles': True}, 'population': {'age_range': [13.0, 54.9], 'force_ovulatory_cycles': False}}	Participant	Investigator-configured study cohorts	Participant age
Menstrual	Cycle length and variability	Both	Age-specific low/high participant-variability mixture initialized by a deterministic surrogate and refined in the full simulator; 17-day-shifted lognormal cycle sampler; broad 18–120-day software bounds	Participant /cycle	Calibrated to Li et al. 2023; Cunningham et al. 2024 held out	Age-specific mean, pooled within-person SD, participant irregularity, short/long tails, and held-out 12-month cross-check

Domain	Parameter	Cohort	Setting / sampling distribution	Level	Source / rationale	Realized target
Menstrual	Bleeding duration	Both	Literature-calibrated	Cycle	Bull et al. 2019	Available in simulator validation suite
Menstrual	Ovulation and phase timing	Cohort-specific	Age/stage/modifier-dependent	Participant /cycle	Bull et al. 2019; simulator literature registry	Ovulatory fraction
Menstrual	Hormone trajectories	Both	LH-aligned daily Stricker medians mapped to ovulation with PCHIP; terminal follicular maturation is not stretched in long cycles; deterministic delayed-emergence or failed-wave E2 geometry	Cycle/day	Stricker et al. 2006; Harlow et al. 2000; Mumford et al. 2012	Independent Anckaert subphase amplitudes plus prespecified daily morphology checks
Menstrual	ILP rule	Population	Maximum P4 on O+5 to O+9 < 5.0 ng/mL; nonovulatory/missing cycles flagged	Cycle	Investigator operationalization; threshold requires biological citation	Audit ILP fraction
Menstrual	PCOS	Population	0.1	Participant	Investigator-configured stress-test rate	Participant prevalence
Menstrual	Peri-menarche	Population	0.6	Participant conditional on age <20	Investigator-configured stress-test rate	Participant prevalence
Menstrual	Perimenopause	Population	0.55	Participant conditional on age ≥ 45	Investigator-configured stress-test rate	Participant prevalence
Menstrual	Dysmenorrhea	Population	0.12	Participant	Investigator-configured stress-test rate	Not retained in participant summary
Menstrual	Contraception/IUD settings	Both	Available but not sampled in completed run	Participant	Simulator literature registry	Not applicable to completed run

Domain	Parameter	Cohort	Setting / sampling distribution	Level	Source / rationale	Realized target
Computational	Compact hormone rendering	Non-audit participants	Consumes the same cycle RNG draws as the full renderer and retains identical cycle structure, ovulation, midluteal progesterone, and ILP status while omitting unused daily concentration objects; full curves retained for validation and the 1% audit sample	Participant /cycle	Result-preserving implementation optimization with automated full-versus-compact equivalence tests	Structural and ILP equivalence; full hormone values in retained audit rows
Observation	Diary duration and start	Both	36 months; diary day 1 sampled uniformly within the first generated menstrual cycle	Participant	Study design	36 months with randomized starting phase
Observation	Missingness	Both	No diary missingness simulated	Day	Study-design simplification	Complete synthetic diaries
Null construction	Independent generators	Both	Separately seeded generators; HORMONE-CYCLE uses SHA-256-separated profile, cycle, and observation-start streams, starts at a uniformly sampled first-cycle day, and aligns directly without wrapping	Participant	Study design	Direct alignment and randomized menstrual starting phase preserved without changing latent profile/cycle draws
Null construction	Master seed	Both	20260505	Run	Reproducibility control	Manifested
Null construction	Phase modes	Both	strict_herzog, luteal_anchored_ovulatory	Analysis	Herzog criteria plus sensitivity	Both modes retained
Null construction	Cohort sizes	Cohort-specific	{'healthy_ovulatory': 50000, 'population': 50000}	Run	Monte Carlo design	100,000 total

3.2 Assumption-by-assumption review

Within this section, menstrual (M), follicular (F), ovulatory (O), and luteal (L) identify cycle phases. Type C1 is the perimenstrual pattern, type C2 is the periovulatory pattern, and type C3 is the inadequate-luteal-phase pattern. Inadequate luteal phase (ILP) is the operational designation defined in Section 3.4.

Appendix Table S2. Assumptions, rationale, and implications for interpretation. The Herzog ratio definitions, thresholds, and inadequate-luteal-phase context derive from Herzog and colleagues (28,29).

Assumption	Rationale	Limitation / implication
Independent generators	Defines the false-positive estimand through independent seizure and hormone random streams.	True catamenial effects and false negatives fall outside this null simulation.
Direct calendar-day alignment	Preserves the separately seeded seizure and hormone outputs while HORMONE-CYCLE selects a uniformly random first-cycle start day.	No end-to-start wraparound is introduced; boundary exposure is randomized across menstrual phases.
36-month diaries	Provides long follow-up and permits shorter nested window comparisons.	Medication changes and secular trends fall outside the data-generating model.
Healthy ovulatory cohort	Provides a lower-heterogeneity reference in which C3 is not applicable.	This reference cohort represents ovulatory menstrual cycling under the configured age range.
Heterogeneous cohort	Stress-tests irregularity, anovulation, and configured modifiers.	Configured modifier rates serve as stress-test inputs; population prevalence estimation falls outside scope.
Strict Herzog phases	Preserves historical comparability with the original criteria.	The ovulatory interval expands in longer cycles; the luteal-anchored mode tests a fixed interval.
Unlabeled days	Matches the historical phase partition and prevents implicit reassignment.	Reduces observed denominators and can make short windows indeterminate.
Exact thresholds 1.69/1.83/1.62	Uses the published distributional inflection points as the implemented thresholds.	Threshold performance remains dependent on sparse numerator and comparator counts.
Zero-comparator handling	Reports a positive numerator over a zero comparator as +infinity and reports 0/0 as undefined.	Makes the sparse-event vulnerability transparent rather than silently adding a continuity correction.
Inadequate-luteal-phase threshold <5 ng/mL	Operationalizes the ovulatory threshold used in the original Herzog phase study.	The cutoff captures one operational definition of luteal adequacy.
Nonovulatory cycles designated inadequate luteal phase	Matches the implemented C3 adapter and the C3 concept of inadequate luteal-phase cycles.	Type C3 sensitivity depends directly on the simulated prevalence of anovulation.
Minimum-data rule	Investigator-selected sensitivity to reduce classifications dominated by very sparse seizure counts.	The four-month or six-cycle and four-seizure-day cutoffs are pragmatic sensitivity settings.
Negative-binomial type C1/type C2 calibration	Checks rejection behavior with a model family concordant with the CHOCOLATES count process.	This analysis is a statistical calibration check for the simulator-based study.
Exploratory negative-binomial type C3	Provides a separately specified follicular-versus-combined-phase contrast in ILP cycles.	Uses only the saved 1% daily audit sample and stricter eligibility.

Assumption	Rationale	Limitation / implication
Randomized diary boundary	Diary day 1 is selected uniformly from the realized days of the first generated menstrual cycle, after which generation proceeds forward.	The first and final observed cycles may be partial; analyses requiring complete cycles exclude them.
Stationary treatment and background risk	Keeps the null stationary and interpretable.	Limits transportability to clinical diaries with treatment changes or disease evolution.

3.3 Menstrual-phase labeling and complete-cycle rules

Phase labels were assigned to each complete menstrual cycle before any analysis-window subset was taken. Forward day counts from menstrual onset, with day 1 as the first bleeding day. Backward day equals the forward day minus the cycle length plus one, so the final cycle day is -1 . Label priority is menstrual, follicular, ovulatory, and then luteal. Days outside the listed ranges remain unlabeled in the strict definition.

Appendix Table S3. Implemented strict Herzog and luteal-anchored sensitivity phase boundaries. Forward day is counted from menstrual onset and backward day from the next onset. Priority rules prevent overlapping labels. The luteal-anchored definition fixes the ovulatory interval immediately before the luteal phase and assigns the remaining nonmenstrual days to the follicular phase.

Phase	Strict Herzog definition	Luteal-anchored sensitivity
Menstrual (M)	Forward days 1–3 or backward days -3 to -1	Same
Follicular (F)	Forward days 4–9	All days not assigned to M, O, or L
Ovulatory (O)	Forward day ≥ 10 and backward day ≤ -13	Backward days -16 to -13
Luteal (L)	Backward days -12 to -4	Backward days -12 to -4
Unlabeled	All remaining days	None

A complete cycle contains every daily row from cycle day 1 through its recorded cycle length. The exact three-cycle Herzog analysis additionally requires three complete cycles, each 23–35 days long. For combined phases, average daily seizure frequency is calculated as the total seizure count across all included phases divided by the total number of labeled days across those phases.

3.4 Type C3 and inadequate-luteal-phase algorithm

Type C3 is the inadequate-luteal-phase catamenial pattern. The present adapter is a custom operationalization. Herzog et al. used a single day-22 progesterone measurement and a 5-nanogram-per-milliliter threshold. (28) This study borrowed that threshold and defined the timing and aggregation algorithm separately.

The adapter designates a nonovulatory cycle or a cycle without a valid ovulation day as inadequate. For an ovulatory cycle, it takes the maximum progesterone concentration from days 5–9 after ovulation. If the interval reaches the diary boundary, it uses the available postovulation days. A maximum below 5 nanograms per milliliter is designated inadequate. A maximum of exactly 5 is therefore adequate in the implemented rule. Complete HORMONE-CYCLE daily outputs contain progesterone on every simulated day. The adapter also contains a defensive missing-field branch that designates the cycle inadequate; that branch was never exercised in this completed simulation and should be treated as an implementation limitation.

Type C3 applicability is evaluated within each analysis window. A pooled window is applicable when it contains at least one designated cycle. The completed study evaluated type C3 only in the heterogeneous cohort as a study-design applicability rule. The healthy cohort was omitted from type C3 reporting; enforced ovulation does not establish that low luteal progesterone is biologically impossible.

3.5 Herzog seizure-frequency ratios and thresholds

Average daily seizure frequency (ADSF) is the number of seizures in a phase divided by the number of labeled days in that phase. Menstrual (M), follicular (F), ovulatory (O), and luteal (L) denote the locally defined phases. The three ratios are shown in Equations (11)–(13). The exact thresholds 1.69, 1.83, and 1.62 correspond to published distributional inflection points. (28,29)

$$C_1 = \frac{ADSF(M)}{ADSF(F + L)} \quad (11)$$

438 where C_1 is the perimenstrual Herzog ratio; $ADSF$ is average daily seizure frequency over the phase
 439 set in parentheses; M is menstrual-phase days; F is follicular-phase days; L is luteal-phase days.

$$C_2 = \frac{ADSF(O)}{ADSF(F + L)} \quad (12)$$

440 where C_2 is the periovulatory Herzog ratio; $ADSF$ is average daily seizure frequency over the phase
 441 set in parentheses; O is ovulatory-phase days; F is follicular-phase days; L is luteal-phase days.

$$C_3 = \frac{ADSF(O + L + M)}{ADSF(F)} \quad (13)$$

442 where C_3 is the inadequate-luteal-phase Herzog ratio; $ADSF$ is average daily seizure frequency over
 443 the phase set in parentheses; O is ovulatory-phase days; L is luteal-phase days; M is menstrual-phase
 444 days; F is follicular-phase days.

445 A small comparator count can produce a large or undefined ratio. Positive infinity is defined and exceeds every
 446 finite threshold. Ratios of zero divided by zero and ratios with missing phase days are undefined.

447 The exact three-cycle analysis requires exactly three complete 23–35-day cycles and classifies a participant
 448 when at least two cycles are positive. (30)

449 **Appendix Table S4. Cumulative distributions of Herzog seizure-frequency ratios in strict three-complete-cycle windows.**
 450 **Panels A–C show simulations; Panel A also shows the NIH Progesterone Trial baseline C1 distribution (n=294) reported**
 451 **by Herzog (2015). Ratio definitions originate from Herzog et al. (1997) (28,29). Simulation denominators at threshold zero**
 452 **include finite and positive-infinite ratios and exclude undefined ratios; the Herzog column uses all 294 participants.**
 453 **Simulation denominators are below 50,000 because zero-over-zero ratios, missing phase days, and eligibility failures are**
 454 **undefined or unavailable. Positive infinity is included at every applicable threshold.**

455 **Panel A — Type C1**

Herzog ratio threshold ($\geq$)	Healthy n	Healthy %	Heterogeneous n	Heterogeneous %	Herzog 2015, % (n=294) (29)
0	43,579	100.00	43,592	100.00	100.00
1	19,691	45.18	19,592	44.94	66.67
1.69	11,943	27.41	11,896	27.29	38.10
2	10,740	24.64	10,770	24.71	34.35
3	6,110	14.02	6,128	14.06	21.43
4	4,695	10.77	4,751	10.90	17.35
5	3,380	7.76	3,492	8.01	14.97
6	3,128	7.18	3,219	7.38	12.24
7	2,915	6.69	3,004	6.89	10.54
8	2,526	5.80	2,619	6.01	8.16

Herzog ratio threshold ($\geq$)	Healthy n	Healthy %	Heterogeneous n	Heterogeneous %	Herzog 2015, % (n=294) (29)
9	2,436	5.59	2,544	5.84	7.48
10	2,269	5.21	2,375	5.45	6.46

456

457 **Panel B — Type C2**

Herzog ratio threshold ($\geq$)	Healthy n	Healthy %	Heterogeneous n	Heterogeneous %
0	43,958	100.00	44,039	100.00
1	19,966	45.42	20,525	46.61
1.83	11,480	26.12	11,679	26.52
2	10,423	23.71	10,656	24.20
3	6,571	14.95	6,780	15.40
4	4,826	10.98	5,099	11.58
5	3,911	8.90	4,168	9.46
6	3,446	7.84	3,700	8.40
7	3,102	7.06	3,387	7.69
8	2,921	6.64	3,165	7.19
9	2,814	6.40	3,048	6.92
10	2,709	6.16	2,936	6.67

458

459 **Panel C — Type C3**

Herzog ratio threshold ($\geq$)	Heterogeneous n	Heterogeneous %
0	16,255	100.00
1	10,124	62.28
1.62	8,667	53.32
2	8,215	50.54
3	7,625	46.91
4	7,388	45.45
5	7,256	44.64
6	7,192	44.24
7	7,147	43.97
8	7,121	43.81

Herzog ratio threshold ($\geq$)	Heterogeneous n	Heterogeneous %
9	7,095	43.65
10	7,078	43.54

460

461 **Panel D — Sparse-comparator audit**

Ratio	Cohort	Finite	Positive infinity	Undefined	At or above threshold	Positive-infinite share
Type C1	Healthy ovulatory	41,602	1,977	6,421	11,943	1,977 (16.6%)
Type C1	Heterogeneous	41,547	2,045	6,408	11,896	2,045 (17.2%)
Type C2	Healthy ovulatory	41,601	2,357	6,042	11,480	2,357 (20.5%)
Type C2	Heterogeneous	41,445	2,594	5,961	11,679	2,594 (22.2%)
Type C3	Heterogeneous	9,281	6,974	3,468	8,667	6,974 (80.5%)

462

463 Panel D audits pooled ratio status in the same strict three-cycle data used for panels A–C. The final exact
 464 Herzog classification separately requires at least two of three individual cycles to be positive. “Positive-infinite
 465 share” is the number of positive-infinite ratios divided by all pooled ratios at or above the applicable threshold.

466 **3.6 Type C3 window-duration sensitivity**

467 Type C3 is the inadequate-luteal-phase pattern. This sensitivity analysis varies the amount of observation
 468 available for that pattern.

469 **Appendix Table S5. Windowed type C3 false-positive classifications in the heterogeneous cohort. All 50,000**
 470 **heterogeneous-cohort participants were attempted. “Applicable” indicates at least one designated inadequate-luteal-phase**
 471 **cycle in the window. “Classifiable” additionally requires a defined ratio. “All-attempted rate” is positive divided by**
 472 **50,000. The healthy ovulatory cohort was omitted under the study-design applicability rule described in Section 3.4.**

Window	Applicable	Classifiable	Positive	Positive among classifiable, % (95% Wilson confidence interval)	All-attempted rate
1 month	16,125	7,694	4,409	57.3% (56.2%–58.4%)	8.8%
3 months	22,097	16,052	8,598	53.6% (52.8%–54.3%)	17.2%
4 months	24,113	18,655	9,724	52.1% (51.4%–52.8%)	19.4%
6 months	26,716	21,817	10,697	49.0% (48.4%–49.7%)	21.4%
9 months	29,699	25,221	11,651	46.2% (45.6%–46.8%)	23.3%
12 months	32,166	27,692	12,190	44.0% (43.4%–44.6%)	24.4%
18 months	35,911	31,392	13,007	41.4% (40.9%–42.0%)	26.0%
24 months	38,869	34,347	13,563	39.5% (39.0%–40.0%)	27.1%
36 months	43,792	38,538	14,296	37.1% (36.6%–37.6%)	28.6%
3 complete cycles	19,723	16,255	8,667	53.3% (52.6%–54.1%)	17.3%
6 complete cycles	25,329	21,796	10,436	47.9% (47.2%–48.5%)	20.9%

Window	Applicable	Classifiable	Positive	Positive among classifiable, % (95% Wilson confidence interval)	All-attempted rate
12 complete cycles	31,069	27,466	11,813	43.0% (42.4%–43.6%)	23.6%
36 months	43,792	38,538	14,296	37.1% (36.6%–37.6%)	28.6%

473

3.7 Investigator-selected minimum-data sensitivity

The minimum-data thresholds are investigator-selected sensitivity settings. They represent the minimum amount of seizure information an investigator is willing to accept before classification. The analysis reports every setting; investigators select the sensitivity appropriate to their scientific question.

Appendix Table S6. Classification across observation windows and minimum seizure-day requirements. Each cohort contains 50,000 attempted participants. “Positive among classifiable” is positive divided by classifiable with a 95% Wilson score confidence interval. “All-attempted rate” is positive divided by 50,000 (31).

Cohort	Window	Minimum seizure days	Classifiable	Positive	Positive among classifiable, % (95% Wilson confidence interval)	All-attempted rate
Healthy ovulatory	4 months	2	43,188	16,441	38.1% (37.6%–38.5%)	32.9%
Healthy ovulatory	6 months	2	45,261	15,050	33.3% (32.8%–33.7%)	30.1%
Healthy ovulatory	12 months	2	47,522	11,586	24.4% (24.0%–24.8%)	23.2%
Healthy ovulatory	36 months	2	49,193	5,454	11.1% (10.8%–11.4%)	10.9%
Heterogeneous	4 months	2	43,102	16,421	38.1% (37.6%–38.6%)	32.8%
Heterogeneous	6 months	2	45,198	15,450	34.2% (33.7%–34.6%)	30.9%
Heterogeneous	12 months	2	47,510	11,792	24.8% (24.4%–25.2%)	23.6%
Heterogeneous	36 months	2	49,168	5,535	11.3% (11.0%–11.5%)	11.1%
Healthy ovulatory	4 months	3	40,269	14,719	36.6% (36.1%–37.0%)	29.4%
Healthy ovulatory	6 months	3	43,129	13,745	31.9% (31.4%–32.3%)	27.5%
Healthy ovulatory	12 months	3	46,362	10,891	23.5% (23.1%–23.9%)	21.8%
Healthy ovulatory	36 months	3	48,777	5,201	10.7% (10.4%–10.9%)	10.4%
Heterogeneous	4 months	3	40,158	14,781	36.8% (36.3%–37.3%)	29.6%
Heterogeneous	6 months	3	43,147	14,273	33.1% (32.6%–33.5%)	28.5%
Heterogeneous	12 months	3	46,394	11,135	24.0% (23.6%–24.4%)	22.3%
Heterogeneous	36 months	3	48,763	5,275	10.8% (10.5%–11.1%)	10.5%
Healthy ovulatory	4 months	4	37,627	13,437	35.7% (35.2%–36.2%)	26.9%

Cohort	Window	Minimum seizure days	Classifiable	Positive	Positive among classifiable, % (95% Wilson confidence interval)	All-attempted rate
Healthy ovulatory	6 months	4	41,140	12,805	31.1% (30.7%–31.6%)	25.6%
Healthy ovulatory	12 months	4	45,268	10,394	23.0% (22.6%–23.4%)	20.8%
Healthy ovulatory	36 months	4	48,359	4,993	10.3% (10.1%–10.6%)	10.0%
Heterogeneous	4 months	4	37,366	13,430	35.9% (35.5%–36.4%)	26.9%
Heterogeneous	6 months	4	41,163	13,289	32.3% (31.8%–32.7%)	26.6%
Heterogeneous	12 months	4	45,278	10,569	23.3% (23.0%–23.7%)	21.1%
Heterogeneous	36 months	4	48,333	5,074	10.5% (10.2%–10.8%)	10.1%
Healthy ovulatory	4 months	6	32,717	11,325	34.6% (34.1%–35.1%)	22.7%
Healthy ovulatory	6 months	6	37,514	11,228	29.9% (29.5%–30.4%)	22.5%
Healthy ovulatory	12 months	6	43,128	9,475	22.0% (21.6%–22.4%)	18.9%
Healthy ovulatory	36 months	6	47,556	4,630	9.7% (9.5%–10.0%)	9.3%
Heterogeneous	4 months	6	32,624	11,326	34.7% (34.2%–35.2%)	22.7%
Heterogeneous	6 months	6	37,492	11,674	31.1% (30.7%–31.6%)	23.3%
Heterogeneous	12 months	6	43,131	9,610	22.3% (21.9%–22.7%)	19.2%
Heterogeneous	36 months	6	47,503	4,746	10.0% (9.7%–10.3%)	9.5%

479

480 3.8 Model-concordant negative-binomial calibration checks

481 The primary model-concordant calibration check uses a log-link negative-binomial generalized linear model for daily seizure counts. Participant-full-
482 diary method-of-moments dispersion is used, with cycle fixed effects when the required complete cycles are available. Type C1 and type C2 one-sided
483 Wald tests are adjusted by the Holm method within participant. (32) Type C1 is the perimenstrual contrast, type C2 is the periovulatory contrast, and type
484 C3 is the inadequate-luteal-phase contrast. The Cyclic Heterogeneous Overdispersed Clustered Open-source L-relationship Adjustable Temporally
485 limited E-diary Simulator (CHOCOLATES) uses an overdispersed count process. (2) Concordance is limited to the count family and log link. The

486 regression does not reproduce CHOCOLATES clustering, multidien rhythms, or participant heterogeneity. The indicator IM equals 1 on menstrual days,
487 the indicator IO equals 1 on ovulatory days, and both equal 0 otherwise.

$$\log(\mu_{icd}) = \beta_0 + \beta_M I_{M,icd} + \beta_O I_{O,icd} + \gamma_c \quad (14)$$

488 where μ_{icd} is the expected seizure count for participant i , cycle c , and day d ; i is the participant index; c is the cycle index; d is the day index; β_0
489 is the intercept; β_M is the menstrual-phase log rate-ratio coefficient; β_O is the ovulatory-phase log rate-ratio coefficient; $I_{M,icd}$ is the menstrual-
490 phase indicator; $I_{O,icd}$ is the ovulatory-phase indicator; γ_c is the fixed effect for cycle c .

491 Follicular and luteal days form the reference category because both phase indicators equal 0 on those days.

492 The exploratory type C3 model restricts the data to complete simulator-designated inadequate-luteal-phase cycles and replaces the two phase indicators
493 with one indicator for combined ovulatory, luteal, and menstrual days. Positivity requires a one-sided P value below .05 and a rate ratio of at least 1.62.
494 The minimum data are four complete designated cycles and four seizure days. A robust Poisson model with heteroskedasticity-consistent covariance type
495 0 is attempted only when the negative-binomial fit raises an exception. Both models use a maximum of 100 iterations. Each daily row has one day of
496 exposure, so no offset is required. Equation (15) defines the participant-specific negative-binomial dispersion.

$$\alpha_i = \begin{cases} 1, & \bar{y}_i \leq 0 \text{ or } s_i^2 \leq \bar{y}_i, \\ \max\left(10^{-6}, \min\left(50, \frac{s_i^2 - \bar{y}_i}{\bar{y}_i^2}\right)\right), & s_i^2 > \bar{y}_i > 0. \end{cases} \quad (15)$$

497 where α_i is the negative-binomial dispersion for participant i ; i is the participant index; $\bar{y}_i$ is the participant-specific mean daily seizure count
498 over the full diary; s_i^2 is the participant-specific sample variance of daily seizure counts.

499 The implemented fallback sets dispersion to 1 when the mean is nonpositive or the sample variance does not exceed the mean. Otherwise, the method-of-
500 moments estimate is bounded below by 10^{-6} and above by 50 for numerical stability.

501 Equation (16) gives the exploratory type C3 model.

$$\log(\mu_{icd}) = \beta_0 + \beta_{OLM} I_{OLM,icd} + \gamma_c \quad (16)$$

502 where μ_{icd} is the expected seizure count for participant i , cycle c , and day d ; i is the participant index; c is the cycle index; d is the day index; β_0
503 is the intercept; β_{OLM} is the log rate-ratio coefficient comparing combined ovulatory, luteal, and menstrual days with follicular days; $I_{OLM,icd}$ is
504 the combined ovulatory-luteal-menstrual phase indicator; γ_c is the fixed effect for cycle c .

505 The retained daily audit sample was selected independently within each cohort without replacement at a 1% fraction. NumPy's default random-number
506 generator used a deterministic 32-bit seed derived from master seed 20260505, the cohort name, and "audit_sample." The heterogeneous-cohort seed was

529110050. The type C3 exploratory analysis attempted all 500 retained heterogeneous participants: 427 had ratio-level type C3 applicability, 197 met regression data requirements, 279 had fewer than four complete inadequate-luteal-phase cycles, and 24 had fewer than four seizure days. All 197 classifiable participants used the negative-binomial fit; no robust-Poisson fallback or regression failure occurred. 12 were positive.

Appendix Table S7. Exploratory type C3 negative-binomial calibration result in the retained 1% daily audit sample. All 500 retained heterogeneous participants were attempted; 427 were ratio-applicable, 197 were regression-classifiable, and 12 were positive. “All-attempted rate” is 12/500. The 95% Wilson interval describes Monte Carlo uncertainty under this configured simulation. No robust-Poisson fallback was used.

Attempted	Ratio type C3 applicable	Negative-binomial classifiable	Positive	Positive among classifiable, % (95% Wilson confidence interval)	All-attempted rate
500	427	197	12	6.1% (3.5%–10.3%)	2.4%

3.9 Mutually exclusive pattern decomposition

Type C1 denotes the perimenstrual pattern, type C2 the periovulatory pattern, and type C3 the inadequate-luteal-phase pattern. Table S8 separates their mutually exclusive combinations.

Appendix Table S8. Mutually exclusive participant-window categories in 36-month strict-Herzog windows. Each cohort contains 50,000 attempted participants. “Windowed Herzog” uses the full 36-month diary. “Minimum-data” adds a minimum of four seizure days. Type C3 follows the heterogeneous-cohort applicability rule. Percentages may differ from 100% by rounding.

Cohort / definition	None	Indeterminate	C1 only	C2 only	C1+C2	C3 only	C3 with C1 and/or C2
Healthy / windowed Herzog	87.9%	0.8%	5.6%	3.4%	2.2%	N/A	N/A
Healthy / minimum-data	86.7%	3.3%	5.1%	2.9%	2.0%	N/A	N/A
Heterogeneous / windowed Herzog	63.6%	0.8%	3.7%	2.1%	1.1%	24.1%	4.5%
Heterogeneous / minimum-data	62.6%	3.3%	3.3%	1.7%	1.0%	23.9%	4.1%

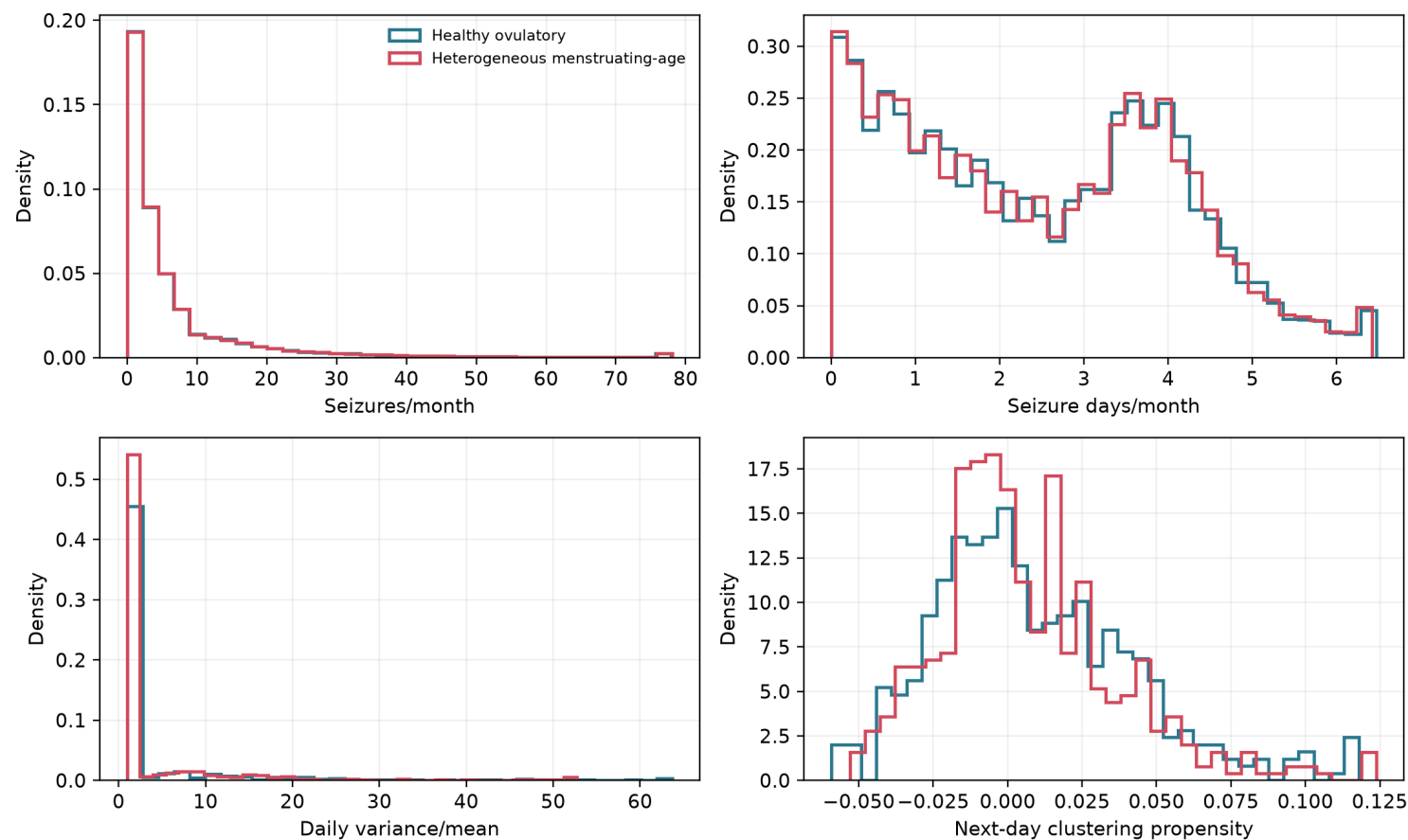
3.10 Simulator-feature associations

The machine-readable file tableS4_simulated_classification_associations.csv reports stratified false-positive classification rates and 95% Wilson score confidence intervals (31) across configured quintiles of age, seizure burden, cycle length and variability, ovulatory fraction, and dominant seizure-cycle period, plus configured binary modifiers. These descriptive results apply to the synthetic study population.

525 **3.11 Realized simulation distributions**

526 The figures in this section show simulated data. Polycystic ovary syndrome (PCOS) and inadequate luteal phase (ILP) are written in abbreviated form
527 within selected panel labels.

Supplementary Figure S1. Realized seizure-process distributions

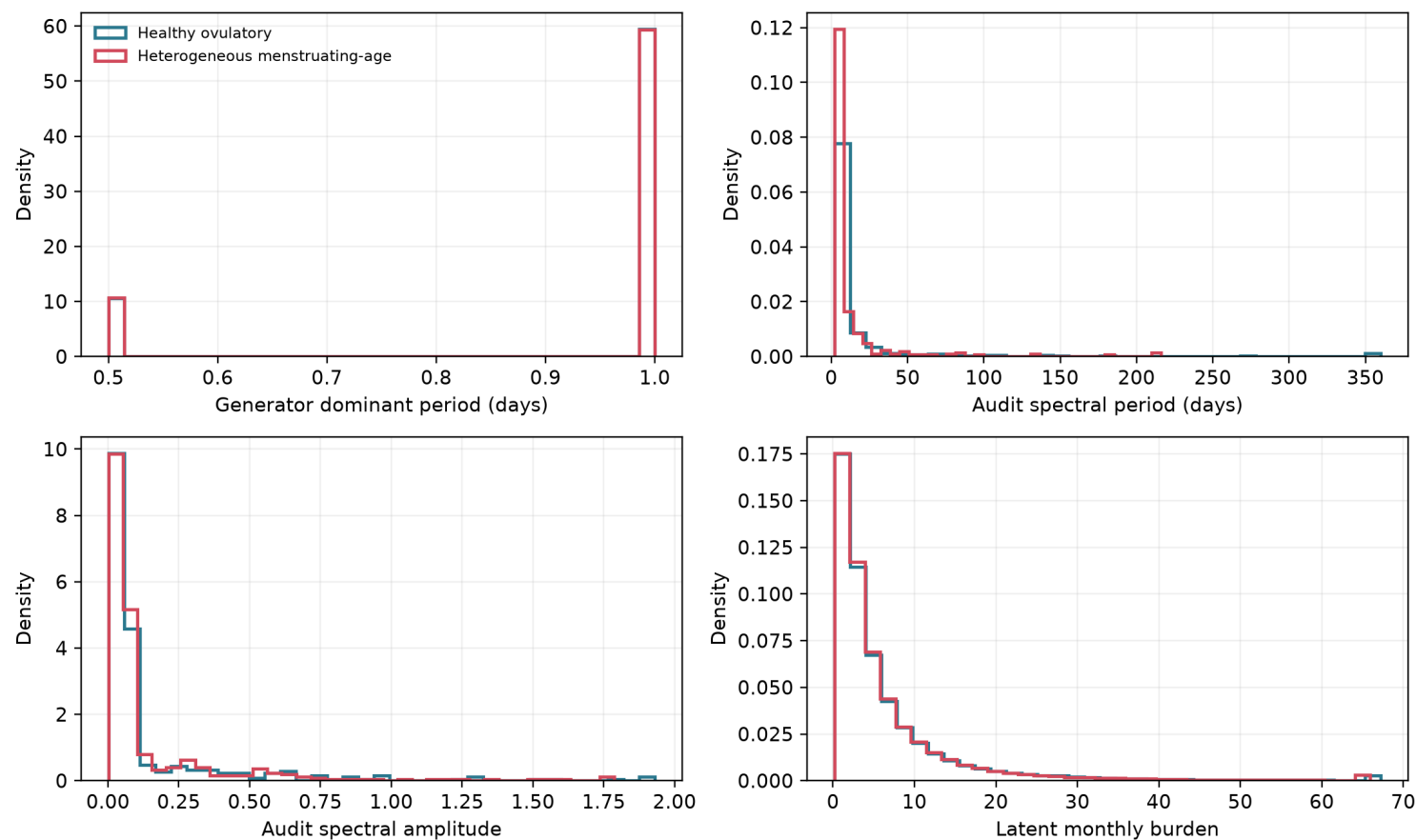


Completed-run distributions; extreme 0.5% tails are winsorized for display. These panels are descriptive calibration outputs, not external empirical validation.

528

529 **Supplementary Figure S1. Realized seizure-process distributions. Seizures per month and seizure days per month use all 50,000 participants in each cohort. Daily count**
 530 **variance divided by mean and next-day clustering propensity use the retained 1% daily audit sample of 500 participants per cohort. All panels display simulated data;**
 531 **monthly measures use 30-day months.**

Supplementary Figure S2. Realized seizure-rhythm distributions



Completed-run distributions; extreme 0.5% tails are winsorized for display. These panels are descriptive calibration outputs, not external empirical validation.

532

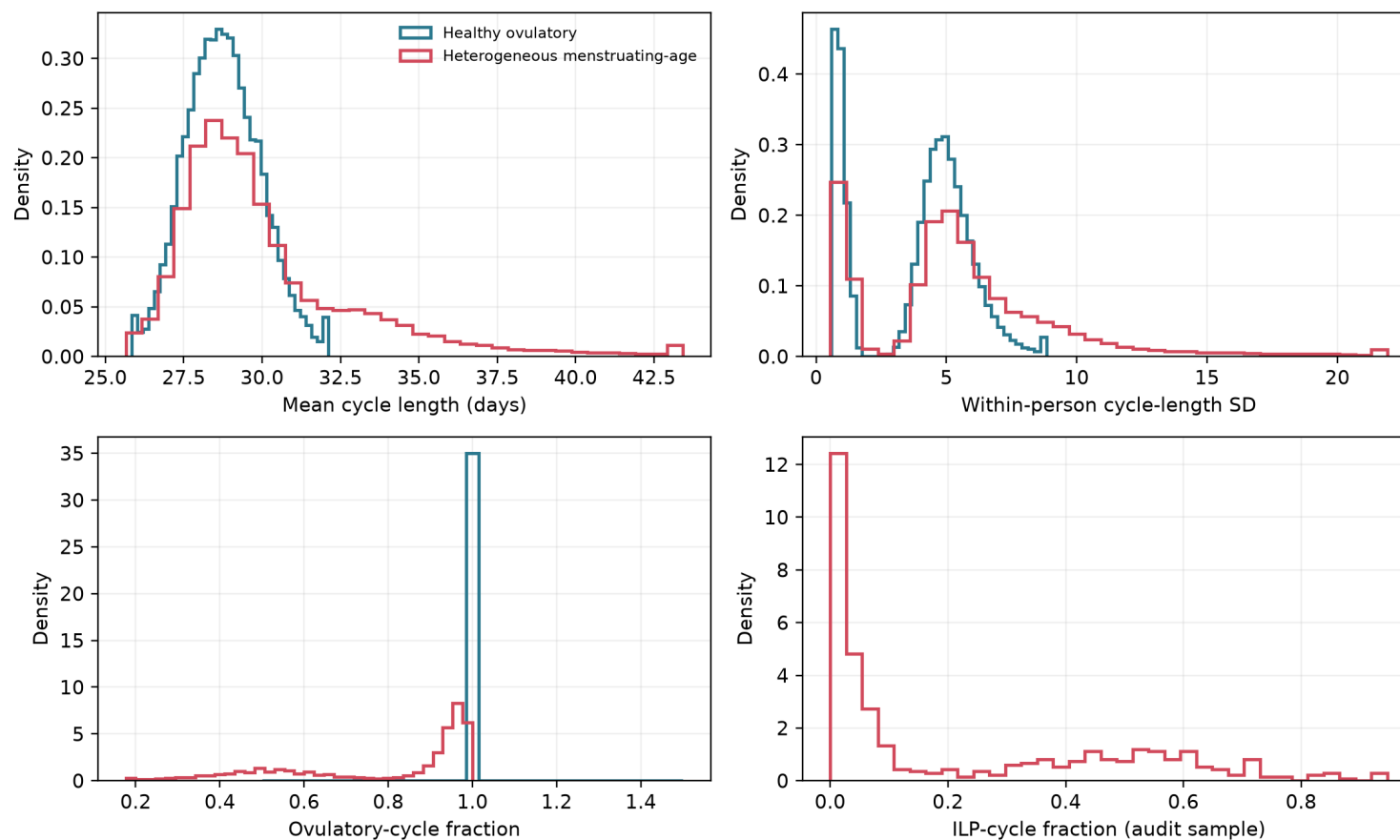
533

534

535

Supplementary Figure S2. Realized seizure-rhythm distributions. Generator-reported dominant period and latent monthly seizure burden use all 50,000 participants per cohort. Spectral period and amplitude use the retained 1% daily audit sample of 500 participants per cohort. Period is in days, spectral amplitude is in seizures per day, and burden is in seizures per 30-day month. All panels display simulated data.

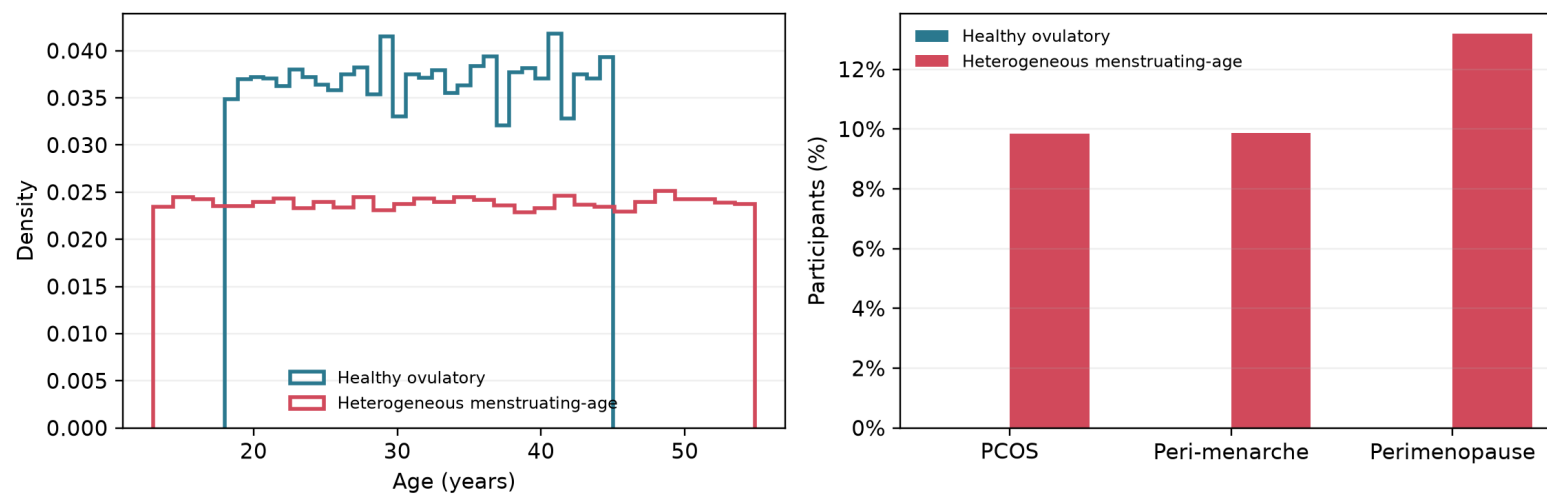
Supplementary Figure S3. Realized menstrual-cycle distributions



Completed-run distributions; extreme 0.5% tails are winsorized for display. These panels are descriptive calibration outputs, not external empirical validation.

Supplementary Figure S3. Realized menstrual-cycle distributions in all 50,000 participants per cohort. Panels show mean cycle length in days, within-participant cycle-length standard deviation in days, ovulatory-cycle fraction, and inadequate luteal phase (ILP) fraction. The healthy cohort was generated with ovulation required. All panels display simulated data.

Supplementary Figure S4. Age and configured medical modifiers

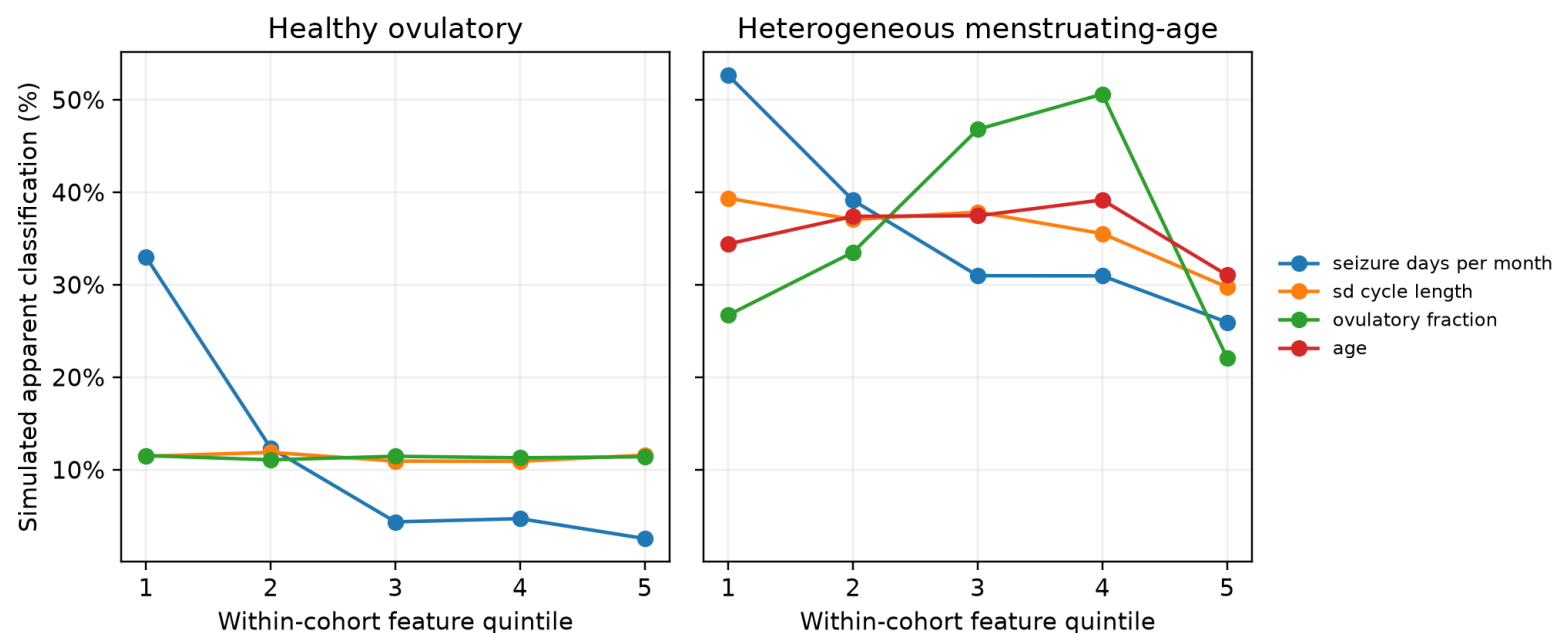


Modifier prevalences are investigator-configured stress-test assumptions, not population estimates.

540

541 **Supplementary Figure S4. Age and configured modifiers among all 50,000 participants per cohort. Polycystic ovary syndrome (PCOS) is abbreviated within the panel.**
 542 **Ages are in years. Modifier frequencies are investigator-configured heterogeneity stress-test settings and do not estimate population prevalence. All panels display**
 543 **simulated data.**

Supplementary Figure S5. Associations with simulated apparent classification



Descriptive associations between simulator inputs/realizations and simulated outcomes; not clinical risk factors or causal effects.

Supplementary Figure S5. Associations with false-positive classification under the independent null model across within-cohort feature quintiles. Each cohort contains 50,000 attempted participants; individual points use the classifiable denominator for the displayed definition. Error bars are 95% Wilson score confidence intervals. The display is descriptive and applies to the synthetic study population.

3.12 Interpretation of the supplementary analyses

The supplementary analyses quantify apparent catamenial classification under independence. They separate the perimenstrual (type C1), periovulatory (type C2), and inadequate-luteal-phase (type C3) patterns; preserve classifiable and all-attempted denominators; and show how diary duration, minimum-data requirements, ratio denominators, and inadequate-luteal-phase prevalence affect the estimates. Medication changes and secular trends are outside the simulation scope and remain limitations for translation to longitudinal clinical diaries.

References

1. Daniel Goldenholz. HORMONE-CYCLE [Python] [Python 3.11+]. GoldenholzLab; 2026. Available from: <https://github.com/GoldenholzLab/catamenial-epilepsy-sim.git>
2. Goldenholz DM, Westover MB. Flexible realistic simulation of seizure occurrence recapitulating statistical properties of seizure diaries. *Epilepsia*. 2023 Feb;64(2):396–405. doi:10.1111/epi.17471
3. Monks T, Currie CSM, Onggo BS, Robinson S, Kunc M, Taylor SJE. Strengthening the reporting of empirical simulation studies: introducing the STRESS guidelines. *J Simul*. 2019;13(1):55–67. doi:10.1080/17477778.2018.1442155
4. Burton A, Altman DG, Royston P, Holder RL. The design of simulation studies in medical statistics. *Stat Med*. 2006 Dec 30;25(24):4279–92. doi:10.1002/sim.2673 PubMed PMID: 16947139.
5. Li H, Gibson EA, Jukic AMZ, Baird DD, Wilcox AJ, Curry CL, et al. Menstrual cycle length variation by demographic characteristics from the Apple Women's Health Study. *NPJ Digit Med*. 2023;6(1):100. Located at: 37248288. doi:10.1038/s41746-023-00848-1
6. Bull JR, Rowland SP, Scherwitzl EB, Scherwitzl R, Danielsson KG, Harper J. Real-world menstrual cycle characteristics of more than 600,000 menstrual cycles. *NPJ Digit Med*. 2019;2:83. Located at: 31482137. doi:10.1038/s41746-019-0152-7
7. Stricker R, Eberhart R, Chevailler MC, Quinn FA, Bischof P, Stricker R. Establishment of detailed reference values for luteinizing hormone, follicle stimulating hormone, estradiol, and progesterone during different phases of the menstrual cycle on the Abbott ARCHITECT analyzer. *Clin Chem Lab Med*. 2006;44(7):883–7. Located at: 16776638. doi:10.1515/CCLM.2006.160
8. Cunningham AC, Pal L, Wickham AP, Prentice C, Goddard FGB, Klepchkova A, et al. Chronicling menstrual cycle patterns across the reproductive lifespan with real-world data. *Scientific Reports*. 2024 May 3;14(1):10172. Located at: 38702411. doi:10.1038/s41598-024-60373-3
9. Harlow SD, Baird DD, Weinberg CR, Wilcox AJ. Urinary oestrogen patterns in long follicular phases. *Hum Reprod*. 2000 Jan;15(1):11–6. doi:10.1093/humrep/15.1.11 PubMed PMID: 10611180.
10. Mumford SL, Steiner AZ, Pollack AZ, Perkins NJ, Filiberto AC, Albert PS, et al. The utility of menstrual cycle length as an indicator of cumulative hormonal exposure. *J Clin Endocrinol Metab*. 2012 Oct;97(10):E1871–1879. doi:10.1210/jc.2012-1350 PubMed PMID: 22837188; PubMed Central PMCID: PMC3674299.
11. Anckaert E, Jank A, Petzold J, Rohsmann F, Paris R, Renggli M, et al. Extensive monitoring of the natural menstrual cycle using the serum biomarkers estradiol, luteinizing hormone and progesterone. *Pract Lab Med*. 2021 May;25:e00211. doi:10.1016/j.plabm.2021.e00211 PubMed PMID: 33869706; PubMed Central PMCID: PMC8042396.
12. Fraser IS, Critchley HOD, Broder M, Munro MG. The FIGO recommendations on terminologies and definitions for normal and abnormal uterine bleeding. *Semin Reprod Med*. 2011 Sep;29(5):383–90. doi:10.1055/s-0031-1287662 PubMed PMID: 22065325.
13. Mortimer R, Asokan G, Baird DD, Wilcox AJ, Abrams K, Curry CL, et al. Variability of menstrual cycles by age, polycystic ovary syndrome, and early-life cycle irregularity in the apple Women's Health Study. *Am J Obstet Gynecol*. 2026 Apr;234(4):1042–69. doi:10.1016/j.ajog.2025.11.031 PubMed PMID: 41297783; PubMed Central PMCID: PMC12915291.
14. Doi SAR, Al-Zaid M, Towers PA, Scott CJ, Al-Shoumer KAS. Irregular cycles and steroid hormones in polycystic ovary syndrome. *Hum Reprod*. 2005 Sep;20(9):2402–8. doi:10.1093/humrep/dei093 PubMed PMID: 15932911.
15. Jarrett BY, Vanden Brink H, Oldfield AL, Lujan ME. Ultrasound Characterization of Disordered Antral Follicle Development in Women with Polycystic Ovary Syndrome. *J Clin Endocrinol Metab*. 2020 Nov 1;105(11):e3847–3861. doi:10.1210/clinem/dgaa515 PubMed PMID: 32785651; PubMed Central PMCID: PMC7473602.
16. Venturoli S, Porcu E, Fabbri R, Paradisi R, Ruggeri S, Bolelli G, et al. Menstrual irregularities in adolescents: hormonal pattern and ovarian morphology. *Horm Res*. 1986;24(4):269–79. doi:10.1159/000180567 PubMed PMID: 3491030.
17. World Health Organization multicenter study on menstrual and ovulatory patterns in adolescent girls. II. Longitudinal study of menstrual patterns in the early postmenarcheal period, duration of bleeding episodes and menstrual cycles. World Health Organization Task Force on Adolescent Reproductive Health. *J Adolesc Health Care*. 1986 Jul;7(4):236–44. PubMed PMID: 3721946.

18. Zhang K, Pollack S, Ghods A, Dicken C, Isaac B, Adel G, et al. Onset of ovulation after menarche in girls: a longitudinal study. *J Clin Endocrinol Metab.* 2008 Apr;93(4):1186–94. doi:10.1210/jc.2007-1846 PubMed PMID: 18252789; PubMed Central PMCID: PMC2291492.
19. Santoro N, Randolph JF. Reproductive hormones and the menopause transition. *Obstet Gynecol Clin North Am.* 2011 Sep;38(3):455–66. doi:10.1016/j.ogc.2011.05.004 PubMed PMID: 21961713; PubMed Central PMCID: PMC3197715.
20. Edelman A, Micks E, Gallo MF, Jensen JT, Grimes DA. Continuous or extended cycle vs. cyclic use of combined hormonal contraceptives for contraception. *Cochrane Database Syst Rev.* 2014 Jul 29;2014(7):CD004695. doi:10.1002/14651858.CD004695.pub3 PubMed PMID: 25072731; PubMed Central PMCID: PMC6837850.
21. Xiao B, Zeng T, Wu S, Sun H, Xiao N. Effect of levonorgestrel-releasing intrauterine device on hormonal profile and menstrual pattern after long-term use. *Contraception.* 1995 Jun;51(6):359–65. doi:10.1016/0010-7824(95)00102-g PubMed PMID: 7554977.
22. Barbosa I, Olsson SE, Odland V, Goncalves T, Coutinho E. Ovarian function after seven years' use of a levonorgestrel IUD. *Adv Contracept.* 1995 Jun;11(2):85–95. doi:10.1007/BF01987274 PubMed PMID: 7491859.
23. Hubacher D, Reyes V, Lillo S, Zepeda A, Chen PL, Croxatto H. Pain from copper intrauterine device insertion: randomized trial of prophylactic ibuprofen. *Am J Obstet Gynecol.* 2006 Nov;195(5):1272–7. doi:10.1016/j.ajog.2006.08.022 PubMed PMID: 17074548.
24. Faundes A, Segal SJ, Adejuwon CA, Brache V, Leon P, Alvarez-Sanchez F. The menstrual cycle in women using an intrauterine device. *Fertil Steril.* 1980 Nov;34(5):427–30. doi:10.1016/s0015-0282(16)45131-9 PubMed PMID: 7439408.
25. Malmqvist R, Petersohn L, Bengtsson LP. Menstrual bleeding with copper-covered intrauterine contraceptive devices. *Contraception.* 1974 Jun;9(6):627–33. doi:10.1016/0010-7824(74)90048-1 PubMed PMID: 4448089.
26. Dawood MY. Primary dysmenorrhea: advances in pathogenesis and management. *Obstet Gynecol.* 2006 Aug;108(2):428–41. doi:10.1097/01.AOG.0000230214.26638.0c PubMed PMID: 16880317.
27. Karoly PJ, Rao VR, Gregg NM, Worrell GA, Bernard C, Cook MJ, et al. Cycles in epilepsy. *Nat Rev Neurol.* 2021;17(5):267–84. Located at: 33723459. doi:10.1038/s41582-021-00464-1
28. Herzog AG, Klein P, Ransil BJ. Three patterns of catamenial epilepsy. *Epilepsia.* 1997;38(10):1082–8. Located at: 9579954. doi:10.1111/j.1528-1157.1997.tb01197.x
29. Herzog AG. Catamenial epilepsy: Update on prevalence, pathophysiology and treatment from the findings of the NIH Progesterone Treatment Trial. *Seizure.* 2015;28:18–25. Located at: 25770028. doi:10.1016/j.seizure.2015.02.024
30. Herzog AG, Harden CL, Liporace J, Pennell P, Schomer DL, Sperling M, et al. Frequency of catamenial seizure exacerbation in women with localization-related epilepsy. *Ann Neurol.* 2004;56(3):431–4. Located at: 15349872. doi:10.1002/ana.20214
31. Newcombe RG. Two-sided confidence intervals for the single proportion: comparison of seven methods. *Stat Med.* 1998 Apr 30;17(8):857–72. doi:10.1002/(sici)1097-0258(19980430)17:8<857::aid-sim777>3.0.co;2-e PubMed PMID: 9595616.
32. Aickin M, Gensler H. Adjusting for multiple testing when reporting research results: the Bonferroni vs Holm methods. *Am J Public Health.* 1996 May;86(5):726–8. doi:10.2105/ajph.86.5.726 PubMed PMID: 8629727; PubMed Central PMCID: PMC1380484.